

Measure Theory
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Introduction: From Riemann to Lebesgue

Measure theory provides a systematic way to assign numerical values to sets and functions. Specifically, we assign a **measure** to a measurable set and an **integral value** to an integrable function. While Riemann integration is a staple of introductory calculus, Lebesgue integration theory represents a significant generalization and completion of those ideas. Lebesgue's framework allows us to assign numbers to a much broader collection of sets and functions than is possible within the Riemann system.

The Shift from Finite to Countable Operations

The fundamental difference between these two theories can be summarized by their relationship with infinity:

- **Riemann Integration** is centered on **finiteness**. It relies on approximations of a finite nature, such as summing the areas of finitely many rectangles to find the area of a bounded subset of $\mathbb{R}^2$. This theory is robust when dealing with finite operations on sets $A_1, \dots, A_n$ or functions $f_1, \dots, f_n$.
- **Lebesgue Integration** is centered on **countable infiniteness**. Riemann integration often fails to behave consistently when faced with operations of a countably infinite nature, such as the union $\bigcup_{n=1}^{\infty} A_n$, the sum $\sum_{n=1}^{\infty} f_n$, or the limit $\lim_{n \rightarrow \infty} f_n$. Lebesgue integration was developed to rectify these disadvantages.

Methodological and Structural Differences

Beyond their handling of infinity, the two theories differ in their execution and scope:

- **Partitioning:** In Riemann's theory, we partition the **domain** of the function, whereas in Lebesgue's theory, we partition the **range**.
- **Generalization:** Riemann's theory is largely restricted to Euclidean space. Conversely, the principles of Lebesgue's theory are applicable to more general spaces, leading to the development of **abstract measure theory**. This abstract approach is highly versatile and intersects with numerous branches of mathematics.

I suggest all students read these two series of notes by **Prof. A. K. Nandakumaran**, Department of Mathematics, IISc Bangalore. These are fantastic articles that provide excellent motivation for studying Measure Theory:

- Meaning of Integration 1
- Meaning of Integration 2

The Vitali Construction: Limits of Generalizing Volume

In this section, we attempt to generalize the concept of volume (specifically length in $\mathbb{R}$) by searching for a set function μ that assigns a "size" to subsets of $\mathbb{R}$. We want this function μ to satisfy certain natural properties:

1. **Normalization:** The length of the unit interval is one, i.e., $\mu([0, 1]) = 1$.
2. **Countable Additivity:** If $E_1, E_2, \dots$ are disjoint subsets, then the measure of their union is the sum of their individual measures: $\mu(\bigcup_{j=1}^{\infty} E_j) = \sum_{j=1}^{\infty} \mu(E_j)$.
3. **Translation Invariance:** If a set E is shifted by a distance r to form a new set $F = E + r$, then $\mu(F) = \mu(E)$.

The following construction, known as the **Vitali Set**, demonstrates that these three assumptions are mutually inconsistent. It proves that there exist sets for which a consistent "length" cannot be defined.

Step-by-Step Construction of the Vitali Set

We work within the interval $[0, 1]$ in $\mathbb{R}$.

Step 1: Define an Equivalence Relation We define a relation $\sim$ on $[0, 1]$. For any $a, b \in [0, 1]$, we say $a \sim b$ if and only if their difference is a rational number:

$$a \sim b \iff a - b \in \mathbb{Q}$$

This partitions $[0, 1]$ into disjoint equivalence classes, where each class contains all points that differ from each other by only a rational value.

Step 2: Form the Vitali Set Y Using the **Axiom of Choice**, we construct a set $Y \subset [0, 1]$ by picking exactly one element from each equivalence class. This set Y is our "Vitali Set."

Step 3: Analyze Rational Translations Let $Q = \mathbb{Q} \cap [-1, 1]$ be the set of rational numbers between -1 and 1 . Since $\mathbb{Q}$ is countable, we can list these rationals as $\{r_1, r_2, r_3, \dots\}$. Consider the translated sets:

$$Y_i = Y + r_i = \{y + r_i : y \in Y\}$$

Step 4: Verify Disjointness We claim that if $r_i \neq r_j$, then $Y_i \cap Y_j = \emptyset$. Suppose there was an element in both. Then $y_1 + r_i = y_2 + r_j$ for some $y_1, y_2 \in Y$. This implies $y_1 - y_2 = r_j - r_i$. Since r_j and r_i are rational, their difference is rational, meaning $y_1 \sim y_2$. However, since Y contains only one element from each class, we must have $y_1 = y_2$, which forces $r_i = r_j$. Thus, the translates are disjoint.

Step 5: Containment and Contradiction Note that since $Y \subset [0, 1]$ and each $r_i \in [-1, 1]$, every translated set Y_i is contained within the interval $[-1, 2]$. Therefore:

$$\bigcup_i Y_i \subset [-1, 2]$$

By translation invariance, all Y_i must have the same measure: $\mu(Y_i) = \mu(Y) = \delta$. By countable additivity:

$$\mu\left(\bigcup_i Y_i\right) = \sum_{i=1}^{\infty} \mu(Y_i) = \sum_{i=1}^{\infty} \delta$$

- If $\delta = 0$, the total measure is 0. But the union of these translates actually covers the interval $[0, 1]$, so the measure must be at least 1. (Contradiction: $0 \geq 1$)
- If $\delta > 0$, the infinite sum $\sum \delta$ diverges to ∞ . But the union is contained in $[-1, 2]$, so the measure cannot exceed 3. (Contradiction: $\infty \leq 3$)

Thus, no such function μ can exist for all subsets of $\mathbb{R}$.

Step 6: Finite Additivity Failure Suppose $\mu(Y) = \delta$. Since $Y \subset [0, 1]$, δ must be some non-negative value.

- If $\delta = 0$, then any countable union of its translates would also have measure 0, which cannot cover $[0, 1]$ (as $\mu([0, 1]) = 1$).
- If $\delta > 0$, we can find a natural number k such that $k\delta > 2$.

Now, choose k distinct rational numbers $q_1, q_2, \dots, q_k$ from $\mathbb{Q} \cap [0, 1]$. Define the translated sets $Y_i = Y + q_i$.

1. **Disjointness:** As shown previously, these translates are disjoint because Y picks only one element per class.
2. **Containment:** Since $Y \subset [0, 1]$ and $q_i \in [0, 1]$, it follows that each $Y_i \subset [0, 2]$. Thus:

$$\bigcup_{i=1}^k Y_i \subset [0, 2]$$

3. **The Contradiction:** By monotonicity and normalization, $\mu(\bigcup_{i=1}^k Y_i) \leq \mu([0, 2]) = 2$. However, by finite additivity and translation invariance:

$$\mu\left(\bigcup_{i=1}^k Y_i\right) = \sum_{i=1}^k \mu(Y_i) = \sum_{i=1}^k \delta = k\delta$$

Since we chose k such that $k\delta > 2$, we get $k\delta \leq 2$, which is a contradiction.

Thus, even finite additivity is inconsistent with translation invariance and normalization.

Remark. The set Y constructed above is an example of a **non-measurable set**. It shows that we cannot assign a meaningful "length" to every possible collection of real numbers if we want to keep the properties of additivity and translation invariance.

Exercise 1. Find two disjoint subsets $A, B \subset [0, 1]$ such that $\mu(A \cup B) \neq \mu(A) + \mu(B)$.

Solution. Consider the translates $Y_1, Y_2, \dots, Y_k$ from the construction above. We know that:

$$\mu(Y_1 \cup Y_2 \cup \dots \cup Y_k) \neq \sum_{j=1}^k \mu(Y_j)$$

We can find the specific pair by induction:

1. Check Y_1 and Y_2 . If $\mu(Y_1 \cup Y_2) \neq \mu(Y_1) + \mu(Y_2)$, we are done. Set $A = Y_1, B = Y_2$.
2. If they are equal, consider the set $(Y_1 \cup Y_2)$ and Y_3 . If $\mu((Y_1 \cup Y_2) \cup Y_3) \neq \mu(Y_1 \cup Y_2) + \mu(Y_3)$, then set $A = Y_1 \cup Y_2$ and $B = Y_3$.
3. Continue this process. We must reach a first natural number $r \leq k$ such that:

$$\mu\left(\left(\bigcup_{j=1}^{r-1} Y_j\right) \cup Y_r\right) \neq \mu\left(\bigcup_{j=1}^{r-1} Y_j\right) + \mu(Y_r)$$

This stage r is guaranteed to exist because if additivity held for every step up to k , the total sum would equal the measure of the union, which we already proved is impossible.

Taking $A = \bigcup_{j=1}^{r-1} Y_j$ and $B = Y_r$ provides the required disjoint sets. □

σ -Algebras and Measurable Spaces

Definition 1. Let X be a nonempty set. An **algebra** of sets on X (also called a **field**) is a nonempty collection $\mathcal{A}$ of subsets of X that satisfies the following closure properties:

1. **Closed under finite unions:** If $E_1, \dots, E_n \in \mathcal{A}$, then $\bigcup_{j=1}^n E_j \in \mathcal{A}$.
2. **Closed under complements:** If $E \in \mathcal{A}$, then $E^c \in \mathcal{A}$.

Definition 2. A σ -**algebra** (or σ -**field**) is an algebra that is also closed under countable unions. That is, if $\{E_j\}_{j=1}^{\infty} \subset \mathcal{A}$, then $\bigcup_{j=1}^{\infty} E_j \in \mathcal{A}$.

Remark. By applying De Morgan's laws, $\bigcap E_j = (\bigcup E_j^c)^c$, we observe that algebras and σ -algebras are also closed under finite and countable intersections, respectively. Furthermore, any algebra $\mathcal{A}$ necessarily contains the empty set $\emptyset$ and the whole set X , since for any $E \in \mathcal{A}$, we have $\emptyset = E \cap E^c$ and $X = E \cup E^c$.

Proposition. An algebra $\mathcal{A}$ is a σ -algebra if and only if it is closed under countable disjoint unions.

Proof. Suppose $\{E_j\}_{j=1}^{\infty} \subset \mathcal{A}$. To show the countable union is in $\mathcal{A}$, we can define a sequence of disjoint sets $\{F_k\}$ as follows:

$$F_1 = E_1, \quad F_k = E_k \setminus \left[\bigcup_{j=1}^{k-1} E_j \right] = E_k \cap \left[\bigcup_{j=1}^{k-1} E_j \right]^c \quad \text{for } k > 1$$

These F_k belong to $\mathcal{A}$ because $\mathcal{A}$ is an algebra (closed under finite unions and complements). By construction, the F_k are disjoint and satisfy $\bigcup_{j=1}^{\infty} E_j = \bigcup_{j=1}^{\infty} F_j$. Thus, if $\mathcal{A}$ is closed under countable disjoint unions, it must also be closed under all countable unions. □

Remark. The technique of replacing an arbitrary sequence of sets with a disjoint sequence such that their union remains unchanged is a fundamental tool in measure theory.

Example 1. Let X be any non-empty set. The following collections of subsets are σ -algebras on X :

1. **The Power Set:** $\mathcal{P}(X)$, the collection of all subsets of X .
2. **The Trivial σ -algebra:** $\{\emptyset, X\}$, the smallest possible σ -algebra.
3. **The Countable or Co-countable σ -algebra:** If X is an uncountable set, then the collection

$$\mathcal{A} = \{E \subset X : E \text{ is countable or } E^c \text{ is countable}\}$$

is a σ -algebra.

Remark. The verification of Example 3 relies on the following logic regarding countable unions:

- **Case 1 (All sets are countable):** If we have a sequence $\{E_j\}_{j=1}^\infty \subset \mathcal{A}$ where every E_j is countable, then their union $\bigcup_{j=1}^\infty E_j$ is a countable union of countable sets, which is itself countable. Thus, the union belongs to $\mathcal{A}$.
- **Case 2 (At least one set is co-countable):** If at least one E_{j_0} is co-countable (meaning $E_{j_0}^c$ is countable), then the complement of the union $(\bigcup E_j)^c = \bigcap E_j^c$ is a subset of $E_{j_0}^c$. Since $E_{j_0}^c$ is countable and any subset of a countable set is countable, the complement of the union is countable. Thus, the union is co-countable and belongs to $\mathcal{A}$.

Exercise 2. Prove that the intersection of any family of σ -algebras on a set X is again a σ -algebra on X .

Solution. Let $\{\mathcal{M}_\alpha\}_{\alpha \in I}$ be a family of σ -algebras on X . Let $\mathcal{M} = \bigcap_{\alpha \in I} \mathcal{M}_\alpha$.

1. Since $X \in \mathcal{M}_\alpha$ for every α , $X \in \mathcal{M}$.
2. If $E \in \mathcal{M}$, then $E \in \mathcal{M}_\alpha$ for all α . Since each $\mathcal{M}_\alpha$ is a σ -algebra, $E^c \in \mathcal{M}_\alpha$ for all α , so $E^c \in \mathcal{M}$.
3. If $\{E_j\}_{j=1}^\infty \subset \mathcal{M}$, then $\{E_j\}_{j=1}^\infty \subset \mathcal{M}_\alpha$ for all α . Thus $\bigcup E_j \in \mathcal{M}_\alpha$ for all α , implying $\bigcup E_j \in \mathcal{M}$.

□

Definition 3. Let $\mathcal{E}$ be any subset of the power set $\mathcal{P}(X)$. There exists a unique smallest σ -algebra containing $\mathcal{E}$, denoted by $\mathcal{M}(\mathcal{E})$. It is defined as the intersection of all σ -algebras on X that contain $\mathcal{E}$:

$$\mathcal{M}(\mathcal{E}) = \bigcap \{\mathcal{M} : \mathcal{M} \text{ is a } \sigma\text{-algebra and } \mathcal{E} \subset \mathcal{M}\}$$

This intersection is well-defined because there is always at least one such σ -algebra, namely $\mathcal{P}(X)$. $\mathcal{M}(\mathcal{E})$ is called the **σ -algebra generated by $\mathcal{E}$** .

Lemma. If $\mathcal{E} \subset \mathcal{M}(\mathcal{F})$, then $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\mathcal{F})$.

Proof. By definition, $\mathcal{M}(\mathcal{E})$ is the *smallest* σ -algebra containing the collection of sets $\mathcal{E}$. This means that if $\mathcal{A}$ is any σ -algebra such that $\mathcal{E} \subset \mathcal{A}$, then it must follow that $\mathcal{M}(\mathcal{E}) \subset \mathcal{A}$.

In this case, we are given that $\mathcal{E} \subset \mathcal{M}(\mathcal{F})$. We already know by definition that $\mathcal{M}(\mathcal{F})$ is a σ -algebra. Therefore, $\mathcal{M}(\mathcal{F})$ is a σ -algebra that contains $\mathcal{E}$. By the minimality of $\mathcal{M}(\mathcal{E})$ discussed above, we conclude that $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\mathcal{F})$. □

Further Definitions in σ -Algebras

Definition 4. Let $\mathcal{E}$ be a collection of subsets of X . The **σ -algebra generated by $\mathcal{E}$** , denoted by $\mathcal{M}(\mathcal{E})$, is the unique smallest σ -algebra containing $\mathcal{E}$. It is defined as the intersection of all σ -algebras on X that contain $\mathcal{E}$. Since the power set $\mathcal{P}(X)$ is always one such σ -algebra, this intersection is well-defined.

Lemma. If $\mathcal{E} \subset \mathcal{M}(\mathcal{F})$, then $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\mathcal{F})$.

Proof. By definition, $\mathcal{M}(\mathcal{F})$ is a σ -algebra. Since it is given that $\mathcal{E} \subset \mathcal{M}(\mathcal{F})$, then $\mathcal{M}(\mathcal{F})$ is one of the σ -algebras included in the intersection that defines $\mathcal{M}(\mathcal{E})$. Because $\mathcal{M}(\mathcal{E})$ is the intersection of all such σ -algebras, it must be a subset of any individual σ -algebra in that collection, specifically $\mathcal{M}(\mathcal{F})$. $\square$

Definition 5. Let X be a topological space (such as a metric space). The σ -algebra generated by the family of all open sets in X is called the **Borel σ -algebra** on X , denoted by $\mathcal{B}_X$. The members of $\mathcal{B}_X$ are called **Borel sets**.

Remark. Since σ -algebras are closed under complements, the Borel σ -algebra is equivalently generated by the family of all closed sets in X . $\mathcal{B}_X$ includes all open sets, closed sets, countable intersections of open sets, and countable unions of closed sets.

Definition 6. Within the hierarchy of Borel sets, there is standard terminology for specific levels:

- A G_δ set is a countable intersection of open sets.
- An F_σ set is a countable union of closed sets.
- A $G_{\delta\sigma}$ set is a countable union of G_δ sets.
- An $F_{\sigma\delta}$ set is a countable intersection of F_σ sets.

The symbols δ and σ originate from the German words *Durchschnitt* (intersection) and *Summe* (union).

Generators of the Borel σ -algebra on $\mathbb{R}$

Proposition. The Borel σ -algebra $\mathcal{B}_{\mathbb{R}}$ is generated by each of the following collections of intervals:

- a. The open intervals: $\mathcal{E}_1 = \{(a, b) : a < b\}$
- b. The closed intervals: $\mathcal{E}_2 = \{[a, b] : a < b\}$
- c. The half-open intervals: $\mathcal{E}_3 = \{(a, b] : a < b\}$ or $\mathcal{E}_4 = \{[a, b) : a < b\}$
- d. The open rays: $\mathcal{E}_5 = \{(a, \infty) : a \in \mathbb{R}\}$ or $\mathcal{E}_6 = \{(-\infty, a) : a \in \mathbb{R}\}$
- e. The closed rays: $\mathcal{E}_7 = \{[a, \infty) : a \in \mathbb{R}\}$ or $\mathcal{E}_8 = \{(-\infty, a] : a \in \mathbb{R}\}$

Proof. Recall that $\mathcal{B}_{\mathbb{R}}$ is generated by the collection of all open sets in $\mathbb{R}$. We will use Lemma 1 to show that $\mathcal{M}(\mathcal{E}_j) = \mathcal{B}_{\mathbb{R}}$ for each j .

Step 1: Establishing $\mathcal{M}(\mathcal{E}_1) = \mathcal{B}_{\mathbb{R}}$ Since every open interval (a, b) is an open set, we have $\mathcal{E}_1 \subset \mathcal{B}_{\mathbb{R}}$, which implies $\mathcal{M}(\mathcal{E}_1) \subset \mathcal{B}_{\mathbb{R}}$. Conversely, every open set in $\mathbb{R}$ can be written as a countable union of open intervals. Since σ -algebras are closed under countable unions, any open set must belong to $\mathcal{M}(\mathcal{E}_1)$. Therefore, $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_1)$, and we conclude $\mathcal{M}(\mathcal{E}_1) = \mathcal{B}_{\mathbb{R}}$.

Step 2: Showing $\mathcal{M}(\mathcal{E}_j) \subset \mathcal{B}_{\mathbb{R}}$ for all j The elements of $\mathcal{E}_j$ for $j \neq 1$ are either open or closed sets, which are inherently Borel sets. For $\mathcal{E}_3$ and $\mathcal{E}_4$, notice that:

$$(a, b] = \bigcap_{n=1}^{\infty} \left(a, b + \frac{1}{n} \right) \quad \text{and} \quad [a, b) = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b \right)$$

Since these are countable intersections of open sets (G_δ sets), they are Borel sets. Thus, $\mathcal{E}_j \subset \mathcal{B}_{\mathbb{R}}$ for all j , which implies $\mathcal{M}(\mathcal{E}_j) \subset \mathcal{B}_{\mathbb{R}}$.

Step 3: Showing $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_j)$ for $j \geq 2$ To prove this, it suffices to show that any open interval (a, b) can be constructed using elements from each $\mathcal{E}_j$.

- **For $\mathcal{E}_2$ (Closed intervals):** We can write $(a, b) = \bigcup_{n=1}^{\infty} [a + \frac{1}{n}, b - \frac{1}{n}]$. Since each $[a + \frac{1}{n}, b - \frac{1}{n}] \in \mathcal{E}_2$, the union is in $\mathcal{M}(\mathcal{E}_2)$.

- **For $\mathcal{E}_3$ (Half-open $(a, b]$):** We can write $(a, b) = \bigcup_{n=1}^{\infty} (a, b - \frac{1}{n}]$. Thus $\mathcal{E}_1 \subset \mathcal{M}(\mathcal{E}_3)$.
- **For $\mathcal{E}_5$ (Open rays (a, ∞)):** Note that $(-\infty, b] = (b, \infty)^c$, so $(-\infty, b] \in \mathcal{M}(\mathcal{E}_5)$. Then $(a, b] = (a, \infty) \cap (-\infty, b] \in \mathcal{M}(\mathcal{E}_5)$. Using the result from $\mathcal{E}_3$, we get $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_5)$.
- **For $\mathcal{E}_7$ (Closed rays $[a, \infty)$):** Note that $(a, \infty) = \bigcup_{n=1}^{\infty} [a + \frac{1}{n}, \infty)$. Thus $\mathcal{E}_5 \subset \mathcal{M}(\mathcal{E}_7)$, which implies $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_7)$.

Similar constructions apply to $\mathcal{E}_4, \mathcal{E}_6$, and $\mathcal{E}_8$ by using symmetry and complements. Since $\mathcal{E}_1 \subset \mathcal{M}(\mathcal{E}_j)$ in all cases, we have $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_j)$. $\square$

Product σ -Algebras

Definition 7. Let $\{X_{\alpha}\}_{\alpha \in A}$ be an indexed collection of nonempty sets, and let $X = \prod_{\alpha \in A} X_{\alpha}$ denote their Cartesian product. For each $\alpha \in A$, let $\pi_{\alpha} : X \rightarrow X_{\alpha}$ be the coordinate maps (or projection maps). If $\mathcal{M}_{\alpha}$ is a σ -algebra on X_{α} for each α , the **product σ -algebra** on X is defined as the σ -algebra generated by the following collection:

$$\{\pi_{\alpha}^{-1}(E_{\alpha}) : E_{\alpha} \in \mathcal{M}_{\alpha}, \alpha \in A\}$$

This σ -algebra is denoted by $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$.

Remark. In the specific case where the index set is finite, such as $A = \{1, \dots, n\}$, we use the notation $\bigotimes_1^n \mathcal{M}_j$ or $\mathcal{M}_1 \otimes \dots \otimes \mathcal{M}_n$. The importance of this definition is primarily established when discussing the measurability of functions on product spaces.

Proposition. If the index set A is countable, then the product σ -algebra $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$ is generated by the collection of measurable rectangles:

$$\mathcal{F} = \left\{ \prod_{\alpha \in A} E_{\alpha} : E_{\alpha} \in \mathcal{M}_{\alpha} \right\}$$

Proof. To prove that $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha} = \mathcal{M}(\mathcal{F})$, we show containment in both directions using Lemma 1.

Step 1: Showing $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha} \subset \mathcal{M}(\mathcal{F})$ Let $\pi_{\alpha}^{-1}(E_{\alpha})$ be a generator of the product σ -algebra, where $E_{\alpha} \in \mathcal{M}_{\alpha}$. This inverse image can be written as a special type of product:

$$\pi_{\alpha}^{-1}(E_{\alpha}) = \prod_{\beta \in A} E_{\beta}, \quad \text{where } E_{\beta} = \begin{cases} E_{\alpha} & \text{if } \beta = \alpha \\ X_{\beta} & \text{if } \beta \neq \alpha \end{cases}$$

Since X_{β} is always in $\mathcal{M}_{\beta}$, this set is a member of the collection $\mathcal{F}$. Thus, every generator of the product σ -algebra is contained in $\mathcal{F}$. By Lemma 1, it follows that $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha} \subset \mathcal{M}(\mathcal{F})$.

Step 2: Showing $\mathcal{M}(\mathcal{F}) \subset \bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$ Now consider an arbitrary element of $\mathcal{F}$, which is a product of the form $\prod_{\alpha \in A} E_{\alpha}$. We can express this product as an intersection of inverse images:

$$\prod_{\alpha \in A} E_{\alpha} = \bigcap_{\alpha \in A} \pi_{\alpha}^{-1}(E_{\alpha})$$

By the definition of the product σ -algebra, each set $\pi_{\alpha}^{-1}(E_{\alpha})$ is a generator and thus belongs to $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$. Since the index set A is countable, this is a **countable intersection** of measurable sets. Since σ -algebras are closed under countable intersections, the resulting product belongs to $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$.

Therefore, $\mathcal{F} \subset \bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$, which by Lemma 1 implies $\mathcal{M}(\mathcal{F}) \subset \bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$. We conclude that the two σ -algebras are identical. $\square$

Proposition. Suppose that for each $\alpha \in A$, the σ -algebra $\mathcal{M}_{\alpha}$ is generated by a collection $\mathcal{E}_{\alpha}$ (i.e., $\mathcal{M}_{\alpha} = \mathcal{M}(\mathcal{E}_{\alpha})$). Then the product σ -algebra $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$ is generated by the collection:

$$\mathcal{F}_1 = \{\pi_{\alpha}^{-1}(E_{\alpha}) : E_{\alpha} \in \mathcal{E}_{\alpha}, \alpha \in A\}$$

Furthermore, if the index set A is countable and $X_{\alpha} \in \mathcal{E}_{\alpha}$ for all α , then $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$ is also generated by the collection of "simple" rectangles:

$$\mathcal{F}_2 = \left\{ \prod_{\alpha \in A} E_{\alpha} : E_{\alpha} \in \mathcal{E}_{\alpha} \right\}$$

Proof. We begin by proving the first assertion regarding $\mathcal{F}_1$.

Step 1: Establishing $\mathcal{M}(\mathcal{F}_1) \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ By definition, each $\mathcal{E}_\alpha \subset \mathcal{M}_\alpha$. Consequently, every set in $\mathcal{F}_1$ is of the form $\pi_\alpha^{-1}(E_\alpha)$ where $E_\alpha \in \mathcal{M}_\alpha$. These are precisely the generators of the product σ -algebra. Since $\mathcal{F}_1 \subset \bigotimes \mathcal{M}_\alpha$, it follows by Lemma 1 that $\mathcal{M}(\mathcal{F}_1) \subset \bigotimes \mathcal{M}_\alpha$.

Step 2: Establishing $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha \subset \mathcal{M}(\mathcal{F}_1)$ For a fixed $\alpha \in A$, consider the collection of subsets of X_α whose inverse images under the coordinate map fall into $\mathcal{M}(\mathcal{F}_1)$:

$$\mathcal{G}_\alpha = \{E \subset X_\alpha : \pi_\alpha^{-1}(E) \in \mathcal{M}(\mathcal{F}_1)\}$$

We can verify that $\mathcal{G}_\alpha$ is a σ -algebra on X_α :

- $X_\alpha \in \mathcal{G}_\alpha$ because $\pi_\alpha^{-1}(X_\alpha) = X$, which is in any σ -algebra.
- If $E \in \mathcal{G}_\alpha$, then $\pi_\alpha^{-1}(E^c) = (\pi_\alpha^{-1}(E))^c \in \mathcal{M}(\mathcal{F}_1)$, so $E^c \in \mathcal{G}_\alpha$.
- If $\{E_j\} \subset \mathcal{G}_\alpha$, then $\pi_\alpha^{-1}(\cup E_j) = \cup \pi_\alpha^{-1}(E_j) \in \mathcal{M}(\mathcal{F}_1)$, so $\cup E_j \in \mathcal{G}_\alpha$.

By the construction of $\mathcal{F}_1$, we know that $\mathcal{E}_\alpha \subset \mathcal{G}_\alpha$. Since $\mathcal{M}_\alpha$ is the *smallest* σ -algebra containing $\mathcal{E}_\alpha$, we must have $\mathcal{M}_\alpha \subset \mathcal{G}_\alpha$. This implies that for *any* $E \in \mathcal{M}_\alpha$, the set $\pi_\alpha^{-1}(E)$ belongs to $\mathcal{M}(\mathcal{F}_1)$. Since these inverse images generate the product σ -algebra, we conclude $\bigotimes \mathcal{M}_\alpha \subset \mathcal{M}(\mathcal{F}_1)$.

Step 3: Proving the second assertion for $\mathcal{F}_2$ If A is countable and $X_\alpha \in \mathcal{E}_\alpha$, we can use the same logic as the previous Proposition. Every element $\pi_\alpha^{-1}(E_\alpha)$ in $\mathcal{F}_1$ can be written as a product $\prod E_\beta$ where $E_\beta = X_\beta$ for $\beta \neq \alpha$. Since $X_\beta \in \mathcal{E}_\beta$, this product is in $\mathcal{F}_2$. Thus $\mathcal{F}_1 \subset \mathcal{F}_2$, and conversely, every element of $\mathcal{F}_2$ is a countable intersection of elements from $\mathcal{F}_1$:

$$\prod_{\alpha \in A} E_\alpha = \bigcap_{\alpha \in A} \pi_\alpha^{-1}(E_\alpha)$$

The result then follows directly from the closure of σ -algebras under countable intersections. □

Borel Sets on Product Spaces

Preliminary: Separability and Product Metrics

Before relating product σ -algebras to Borel σ -algebras, we recall some essential concepts from the theory of metric spaces.

Definition 8. A metric space (X, ρ) is called **separable** if it contains a countable dense subset. That is, there exists a set $C = \{c_1, c_2, \dots\}$ such that for every $x \in X$ and every $\epsilon > 0$, the open ball $B(x, \epsilon)$ contains at least one point of C .

Example 2. The space $\mathbb{R}^n$ with the usual Euclidean metric is separable. The set $\mathbb{Q}^n = \{(q_1, \dots, q_n) : q_i \in \mathbb{Q}\}$ is a countable dense subset of $\mathbb{R}^n$.

Remark. An important property of separable metric spaces is that they possess a **countable base**. Let C be a countable dense subset of X . Let $\mathcal{E}$ be the collection of all open balls with centers in C and rational radii:

$$\mathcal{E} = \{B(c, r) : c \in C, r \in \mathbb{Q}^+\}$$

Since C and $\mathbb{Q}^+$ are countable, $\mathcal{E}$ is a countable collection. Every open set $V \subset X$ can be written as a countable union of members of $\mathcal{E}$. *Proof idea:* For any $x \in V$, there is a ball $B(x, \epsilon) \subseteq V$. By density, we can find $c_i \in C$ very close to x and a rational radius r such that $x \in B(c_i, r) \subseteq B(x, \epsilon) \subseteq V$. Thus, V is the union of all such balls from $\mathcal{E}$ contained within it.

Definition 9. Let $(X_1, \rho_1), \dots, (X_n, \rho_n)$ be metric spaces. The **product metric** ρ on $X = \prod_{j=1}^n X_j$ is typically defined as:

$$\rho((x_1, \dots, x_n), (y_1, \dots, y_n)) = \max_{1 \leq j \leq n} \{\rho_j(x_j, y_j)\}$$

Under this metric, an open ball in X is simply the product of open balls in the component spaces:

$$B_X(x, r) = B_{X_1}(x_1, r) \times B_{X_2}(x_2, r) \times \dots \times B_{X_n}(x_n, r)$$

Furthermore, if each (X_j, ρ_j) is separable with countable dense subset C_j , then X is separable with countable dense subset $C = \prod_{j=1}^n C_j$.

The Relationship between $\mathcal{B}_X$ and $\bigotimes \mathcal{B}_{X_j}$

Proposition. Let $X_1, \dots, X_n$ be metric spaces and let $X = \prod_1^n X_j$ be equipped with the product metric. Then $\bigotimes_1^n \mathcal{B}_{X_j} \subset \mathcal{B}_X$. If each X_j is separable, then $\bigotimes_1^n \mathcal{B}_{X_j} = \mathcal{B}_X$.

Proof. We approach the proof in steps to distinguish between the general case and the case requiring separability.

Step 1: Understanding the Two σ -Algebras We are comparing two distinct ways of constructing a σ -algebra on the product space X :

- **The Borel σ -algebra $\mathcal{B}_X$:** This is generated by the collection of all open sets in the product metric space X . This topology is equivalent to the product topology.
- **The Product σ -algebra $\bigotimes \mathcal{B}_{X_j}$:** This is generated by "measurable cylinders" of the form $\pi_j^{-1}(E_j)$, where $E_j \in \mathcal{B}_{X_j}$.

Step 2: Proving the General Inclusion $\bigotimes_1^n \mathcal{B}_{X_j} \subset \mathcal{B}_X$ By our previous results, the product σ -algebra $\bigotimes \mathcal{B}_{X_j}$ is generated by the collection:

$$\mathcal{F} = \{\pi_j^{-1}(U_j) : U_j \text{ is open in } X_j, 1 \leq j \leq n\}$$

For each j , the projection map π_j is continuous. Therefore, if U_j is an open set in X_j , its inverse image $\pi_j^{-1}(U_j)$ is an open set in the product space X . Since $\mathcal{B}_X$ is the σ -algebra generated by *all* open sets in X , it must contain this sub-collection of open sets. By the minimality of generated σ -algebras (Lemma 1), we conclude that $\bigotimes \mathcal{B}_{X_j} \subset \mathcal{B}_X$.

Step 3: Leveraging Separability in X_j Suppose each X_j is separable. Let C_j be a countable dense subset of X_j . Let $\mathcal{E}_j$ be the countable collection of open balls in X_j with centers in C_j and rational radii. As noted in our preliminary remarks, every open set in X_j is a countable union of balls from $\mathcal{E}_j$. This implies that the Borel σ -algebra $\mathcal{B}_{X_j}$ is generated by the countable collection $\mathcal{E}_j$.

Step 4: Completion of the Equality via Separability Now consider the product space X . Since X is a product of finitely many separable spaces, it is also separable with a countable dense subset $C = \prod C_j$.

1. **Generation of $\mathcal{B}_X$:** Because X is separable, $\mathcal{B}_X$ is generated by the collection of all "product balls":

$$\mathcal{F}_2 = \left\{ \prod_{j=1}^n E_j : E_j \in \mathcal{E}_j \right\}$$

This is because any open set in X is a countable union of such product balls.

2. **Generation of the Product σ -algebra:** From our previous Proposition regarding generators, since $X_j \in \mathcal{E}_j$ (by taking a sufficiently large rational radius) and the index set is finite (and thus countable), the product σ -algebra $\bigotimes \mathcal{B}_{X_j}$ is also generated by the same collection of "simple" rectangles $\mathcal{F}_2$.

Since both $\mathcal{B}_X$ and $\bigotimes \mathcal{B}_{X_j}$ are generated by the exact same collection of sets $\mathcal{F}_2$, they must be identical. $\square$

Remark. The assumption of separability is crucial. In non-separable spaces, the Borel σ -algebra $\mathcal{B}_X$ can be strictly larger than the product σ -algebra because there may be open sets in X that cannot be represented as countable unions of measurable rectangles.

Corollary 1. $\mathcal{B}_{\mathbb{R}^n} = \bigotimes_1^n \mathcal{B}_{\mathbb{R}}$.

Proof. This follows directly from the previous Proposition. Since $\mathbb{R}$ is a separable metric space (containing the countable dense subset $\mathbb{Q}$), its n -fold Cartesian product $\mathbb{R}^n$ satisfies the condition for the Borel σ -algebra to coincide with the product σ -algebra. $\square$

Elementary Family

Definition 10. A collection $\mathcal{E}$ of subsets of X is called an **elementary family** if it satisfies the following three properties:

1. $\emptyset \in \mathcal{E}$.
2. If $E, F \in \mathcal{E}$, then $E \cap F \in \mathcal{E}$ (closed under finite intersections).
3. If $E \in \mathcal{E}$, then its complement E^c is a finite disjoint union of members of $\mathcal{E}$. That is, $E^c = \bigsqcup_{j=1}^n E_j$ for some $E_1, \dots, E_n \in \mathcal{E}$.

Proposition. If $\mathcal{E}$ is an elementary family, the collection $\mathcal{A}$ of finite disjoint unions of members of $\mathcal{E}$ is an algebra.

Proof. To prove that $\mathcal{A}$ is an algebra, we must show that it is closed under finite unions and complements. Recall that an element of $\mathcal{A}$ is of the form $\bigsqcup_{i=1}^n E_i$ where each $E_i \in \mathcal{E}$.

Step 1: Closed under Finite Unions First, we observe that if we can show the union of two elements of $\mathcal{E}$ belongs to $\mathcal{A}$, the general case for elements of $\mathcal{A}$ follows.

Let $A, B \in \mathcal{E}$. We want to show $A \cup B \in \mathcal{A}$. Since $\mathcal{E}$ is an elementary family, the complement B^c can be written as a finite disjoint union of members of $\mathcal{E}$, say $B^c = \bigsqcup_{j=1}^J C_j$ where $C_j \in \mathcal{E}$.

- We can write the set difference as $A \setminus B = A \cap B^c = A \cap (\bigsqcup_{j=1}^J C_j) = \bigsqcup_{j=1}^J (A \cap C_j)$.
- Since $\mathcal{E}$ is closed under finite intersections, each $(A \cap C_j) \in \mathcal{E}$. Because the C_j are disjoint, these intersections are also disjoint. Thus, $A \setminus B \in \mathcal{A}$.
- Now, we can write the union as $A \cup B = (A \setminus B) \sqcup B$. Since both $(A \setminus B)$ and B are finite disjoint unions of elements of $\mathcal{E}$, their union is also a finite disjoint union of elements of $\mathcal{E}$. Hence, $A \cup B \in \mathcal{A}$.

By induction, it follows that if $A_1, \dots, A_n \in \mathcal{E}$, then $\bigcup_1^n A_j \in \mathcal{A}$. Specifically, for the inductive step, we assume the union of $n - 1$ sets is in $\mathcal{A}$ and write:

$$\bigcup_{j=1}^n A_j = A_n \cup \bigcup_{j=1}^{n-1} (A_j \setminus A_n)$$

This expresses the union as a disjoint union of elements in $\mathcal{A}$, confirming that $\mathcal{A}$ is closed under finite unions.

Step 2: Closed under Complements Let $A \in \mathcal{A}$. By definition, $A = \bigcup_{m=1}^n A_m$ where $A_m \in \mathcal{E}$ and the A_m are disjoint. We need to show $A^c \in \mathcal{A}$. Using De Morgan's Laws:

$$A^c = \left(\bigcup_{m=1}^n A_m \right)^c = \bigcap_{m=1}^n A_m^c$$

Since $\mathcal{E}$ is an elementary family, each A_m^c is a finite disjoint union of members of $\mathcal{E}$. Let $A_m^c = \bigsqcup_{j=1}^{J_m} B_m^j$, where $B_m^j \in \mathcal{E}$. Substituting this into the intersection gives:

$$A^c = \bigcap_{m=1}^n \left(\bigcup_{j=1}^{J_m} B_m^j \right)$$

By the distributive law for sets, this intersection of unions can be rewritten as a union of intersections:

$$A^c = \bigcup \{ B_1^{j_1} \cap B_2^{j_2} \cap \dots \cap B_n^{j_n} : 1 \leq j_m \leq J_m \text{ for each } m \}$$

Each term $B_1^{j_1} \cap \dots \cap B_n^{j_n}$ is an intersection of elements of $\mathcal{E}$, so it belongs to $\mathcal{E}$. Furthermore, these terms are disjoint because if we change any index j_m , we are intersecting with a disjoint set from the m -th family. Thus, A^c is a finite disjoint union of members of $\mathcal{E}$, meaning $A^c \in \mathcal{A}$. $\square$

Example 3. Let $\mathcal{E}$ be the collection of all half-open intervals of the form $(a, b]$ in $\mathbb{R}$, including $(a, a] = \emptyset$ and infinite intervals like (a, ∞) and $(-\infty, b]$. This collection forms an **elementary family** because:

1. $\emptyset \in \mathcal{E}$ (by setting $a = b$).
2. The intersection of any two such intervals $(a, b] \cap (c, d] = (\max(a, c), \min(b, d)]$ is either another half-open interval or empty.
3. The complement of any interval $(a, b]^c = (-\infty, a] \cup (b, \infty)$ is a finite disjoint union of two elements from $\mathcal{E}$.

Measures and Measure Spaces

Definition 11. Let X be a set equipped with a σ -algebra $\mathcal{M}$. A **measure** on $\mathcal{M}$ (or on $(X, \mathcal{M})$, or simply on X if $\mathcal{M}$ is understood) is a function $\mu : \mathcal{M} \rightarrow [0, \infty]$ that satisfies the following two properties:

- i. **Null Empty Set:** $\mu(\emptyset) = 0$.
- ii. **Countable Additivity:** If $\{E_j\}_{j=1}^{\infty}$ is a sequence of disjoint sets in $\mathcal{M}$, then:

$$\mu \left(\bigcup_{j=1}^{\infty} E_j \right) = \sum_{j=1}^{\infty} \mu(E_j)$$

Remark. Property (ii) implies **finite additivity**. If $E_1, \dots, E_n$ are disjoint sets in $\mathcal{M}$, then:

$$\mu \left(\bigcup_{j=1}^n E_j \right) = \sum_{j=1}^n \mu(E_j)$$

This is easily seen by taking $E_j = \emptyset$ for all $j > n$. A function μ that satisfies property (i) and finite additivity, but not necessarily countable additivity, is called a **finitely additive measure**.

Definition 12. We use the following standard terminology for the structures involving measures:

- The pair $(X, \mathcal{M})$ is called a **measurable space**.
- The elements of the σ -algebra $\mathcal{M}$ are called **measurable sets**.
- The triple $(X, \mathcal{M}, \mu)$ is called a **measure space**.

Classification of Measures by Size

Let $(X, \mathcal{M}, \mu)$ be a measure space. We classify the measure μ (and sets within $\mathcal{M}$) based on their "size" as follows:

Definition 13. A measure μ is called **finite** if $\mu(X) < \infty$. Note: Since $\mu(X) = \mu(E) + \mu(E^c)$ for any $E \in \mathcal{M}$, the finiteness of $\mu(X)$ implies that $\mu(E) < \infty$ for all measurable sets E .

Definition 14. A measure μ is called **σ -finite** if X can be written as a countable union of measurable sets with finite measure. That is, $X = \bigcup_{j=1}^{\infty} E_j$ where $E_j \in \mathcal{M}$ and $\mu(E_j) < \infty$ for all j .

Definition 15. A set $E \in \mathcal{M}$ is said to be **σ -finite for μ** if $E = \bigcup_{j=1}^{\infty} E_j$ where $E_j \in \mathcal{M}$ and $\mu(E_j) < \infty$ for all j .

Definition 16. A measure μ is called **semifinite** if for every $E \in \mathcal{M}$ with $\mu(E) = \infty$, there exists a measurable subset $F \subset E$ such that $0 < \mu(F) < \infty$.

Theorem 2. Let $(X, \mathcal{M}, \mu)$ be a measure space.

- a. (**Monotonicity**) If $E, F \in \mathcal{M}$ and $E \subset F$, then $\mu(E) \leq \mu(F)$.
- b. (**Subadditivity**) If $\{E_j\}_{j=1}^{\infty} \subset \mathcal{M}$, then $\mu(\bigcup_{j=1}^{\infty} E_j) \leq \sum_{j=1}^{\infty} \mu(E_j)$.
- c. (**Continuity from below**) If $\{E_j\}_{j=1}^{\infty} \subset \mathcal{M}$ and $E_1 \subset E_2 \subset \dots$, then:

$$\mu \left(\bigcup_{j=1}^{\infty} E_j \right) = \lim_{j \rightarrow \infty} \mu(E_j)$$

d. (**Continuity from above**) If $\{E_j\}_{j=1}^\infty \subset \mathcal{M}$, $E_1 \supset E_2 \supset \dots$, and $\mu(E_1) < \infty$, then:

$$\mu\left(\bigcap_{j=1}^\infty E_j\right) = \lim_{j \rightarrow \infty} \mu(E_j)$$

Proof. **(a) Proof of Monotonicity:** Suppose $E, F \in \mathcal{M}$ with $E \subset F$. We can decompose F into two disjoint sets: E and the set difference $F \setminus E$. Thus, $F = E \cup (F \setminus E)$. By the finite additivity of measures:

$$\mu(F) = \mu(E) + \mu(F \setminus E)$$

Since the range of μ is $[0, \infty]$, we know $\mu(F \setminus E) \geq 0$. Therefore, $\mu(F) \geq \mu(E)$.

(b) Proof of Subadditivity: To use the additivity property of measures on a sequence that might not be disjoint, we must first "disjointify" it. Let $\{E_j\}_{j=1}^\infty$ be any sequence in $\mathcal{M}$. Define:

$$F_1 = E_1, \quad F_k = E_k \setminus \left(\bigcup_{j=1}^{k-1} E_j\right) \text{ for } k > 1$$

These F_k are disjoint by construction, and their union is the same as the union of the E_j 's (i.e., $\bigcup_{j=1}^\infty F_j = \bigcup_{j=1}^\infty E_j$). Furthermore, for each j , $F_j \subset E_j$. By countable additivity and the monotonicity proved in (a):

$$\mu\left(\bigcup_{j=1}^\infty E_j\right) = \mu\left(\bigcup_{j=1}^\infty F_j\right) = \sum_{j=1}^\infty \mu(F_j) \leq \sum_{j=1}^\infty \mu(E_j)$$

(c) Proof of Continuity from Below: Let $E_1 \subset E_2 \subset \dots$ be an increasing sequence of sets. We define $E_0 = \emptyset$. We can represent the union as a disjoint union of "rings":

$$\bigcup_{j=1}^\infty E_j = \bigcup_{j=1}^\infty (E_j \setminus E_{j-1})$$

Applying countable additivity:

$$\mu\left(\bigcup_{j=1}^\infty E_j\right) = \sum_{j=1}^\infty \mu(E_j \setminus E_{j-1})$$

By the definition of an infinite series, this is the limit of partial sums:

$$\mu\left(\bigcup_{j=1}^\infty E_j\right) = \lim_{n \rightarrow \infty} \sum_{j=1}^n \mu(E_j \setminus E_{j-1})$$

Since the sets are nested, the finite sum $\sum_{j=1}^n \mu(E_j \setminus E_{j-1})$ is a telescoping sum that simplifies to $\mu(E_n)$. Thus:

$$\mu\left(\bigcup_{j=1}^\infty E_j\right) = \lim_{n \rightarrow \infty} \mu(E_n)$$

(d) Proof of Continuity from Above: Let $E_1 \supset E_2 \supset \dots$ with $\mu(E_1) < \infty$. We transform this into an increasing sequence to apply part (c). Let $F_j = E_1 \setminus E_j$. Since the E_j are decreasing, the F_j are increasing ($F_1 \subset F_2 \subset \dots$). Note that $\bigcup F_j = E_1 \setminus (\bigcap E_j)$. By part (c):

$$\mu(E_1 \setminus \bigcap E_j) = \lim_{j \rightarrow \infty} \mu(F_j)$$

Using the property $\mu(A \setminus B) = \mu(A) - \mu(B)$ (valid here since $\mu(E_1) < \infty$):

$$\mu(E_1) - \mu\left(\bigcap_{j=1}^\infty E_j\right) = \lim_{j \rightarrow \infty} [\mu(E_1) - \mu(E_j)]$$

Because $\mu(E_1)$ is finite, we can subtract it from both sides of the equation:

$$-\mu\left(\bigcap_{j=1}^\infty E_j\right) = -\lim_{j \rightarrow \infty} \mu(E_j)$$

Multiplying by -1 gives the desired result: $\mu(\bigcap_{j=1}^\infty E_j) = \lim_{j \rightarrow \infty} \mu(E_j)$. □

Remark. The condition $\mu(E_1) < \infty$ in part (d) of Theorem 1.8 is a sufficient but not strictly necessary condition for the first term of the sequence; it could be replaced by $\mu(E_n) < \infty$ for any $n > 1$. This is because the first $n - 1$ sets in the sequence do not affect the ultimate intersection $\bigcap_{j=1}^{\infty} E_j$.

However, **some** finiteness assumption is absolutely necessary. It is possible for $\mu(E_j) = \infty$ for all j , yet $\mu(\bigcap_{j=1}^{\infty} E_j) < \infty$. In such cases, the limit of the measures may not equal the measure of the intersection.

Example 4 (Failure of Continuity from Above without Finiteness). Let μ be the **counting measure** on the measurable space $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$. Define a decreasing sequence of sets as follows:

$$E_j = \{n \in \mathbb{N} : n \geq j\} = \{j, j+1, j+2, \dots\}$$

- For every $j \in \mathbb{N}$, the set E_j contains infinitely many elements, so $\mu(E_j) = \infty$. This implies:

$$\lim_{j \rightarrow \infty} \mu(E_j) = \infty$$

- However, the intersection of all these sets is empty: $\bigcap_{j=1}^{\infty} E_j = \emptyset$.
- Therefore, $\mu(\bigcap_{j=1}^{\infty} E_j) = \mu(\emptyset) = 0$.

In this case, $0 \neq \infty$, demonstrating that the property fails when the finiteness condition is violated.

Remark. Every σ -finite measure is semifinite (Exercise 1.13), but not conversely. Most measures that arise in practice are σ -finite, which is fortunate since non- σ -finite measures tend to exhibit pathological behavior.

Exercise 3 (Exercise 1.13). Prove that every σ -finite measure is semifinite.

Proof. Let $(X, \mathcal{M}, \mu)$ be a σ -finite measure space. By definition, $X = \bigcup_{n=1}^{\infty} E_n$ where $E_n \in \mathcal{M}$ and $\mu(E_n) < \infty$ for all n . To show μ is semifinite, let $E \in \mathcal{M}$ be such that $\mu(E) = \infty$. We must find a measurable subset $F \subset E$ with $0 < \mu(F) < \infty$.

- We can decompose E using the sequence $\{E_n\}$:

$$E = E \cap X = E \cap \left(\bigcup_{n=1}^{\infty} E_n \right) = \bigcup_{n=1}^{\infty} (E \cap E_n)$$

- By subadditivity, $\mu(E) \leq \sum_{n=1}^{\infty} \mu(E \cap E_n)$. Since $\mu(E) = \infty$, the sum $\sum \mu(E \cap E_n)$ must also be infinite.
- This implies that there must exist at least one index k for which $\mu(E \cap E_k) > 0$.
- Furthermore, since $E \cap E_k \subseteq E_k$, by monotonicity we have $\mu(E \cap E_k) \leq \mu(E_k) < \infty$.
- Thus, $F = E \cap E_k$ is a measurable subset of E that satisfies $0 < \mu(F) < \infty$.

Therefore, μ is semifinite. □

Measures Defined by Functions

A common way to construct measures on a set X is by using a non-negative function. Let X be any nonempty set, and let $\mathcal{M} = \mathcal{P}(X)$ be its power set. Let $f : X \rightarrow [0, \infty]$ be any function.

Definition 17. The function f determines a measure μ on $\mathcal{M}$ defined by the formula:

$$\mu(E) = \sum_{x \in E} f(x)$$

For the definition of such possibly uncountable sums, one typically defines $\sum_{x \in E} f(x) = \sup\{\sum_{x \in F} f(x) : F \subseteq E, F \text{ is finite}\}$.

Remark. The properties of the measure μ are directly related to the behavior of the function f :

- μ is **semifinite** if and only if $f(x) < \infty$ for every $x \in X$.
- μ is **σ -finite** if and only if μ is semifinite and the set $\{x : f(x) > 0\}$ is countable.

Special Cases of Significance

Two specific instances of this construction are used throughout measure theory and probability:

1. **Counting Measure:** If $f(x) = 1$ for all $x \in X$, then μ is called the **counting measure**.

$$\mu(E) = \begin{cases} |E| & \text{if } E \text{ is finite} \\ \infty & \text{if } E \text{ is infinite} \end{cases}$$

Note that the counting measure is σ -finite if and only if X is a countable set.

2. **Dirac Measure (Point Mass):** If, for some fixed $x_0 \in X$, the function f is defined by $f(x_0) = 1$ and $f(x) = 0$ for $x \neq x_0$, then μ is called the **point mass** or **Dirac measure** at x_0 , often denoted by δ_{x_0} .

$$\delta_{x_0}(E) = \begin{cases} 1 & \text{if } x_0 \in E \\ 0 & \text{if } x_0 \notin E \end{cases}$$

The Dirac measure is always a finite measure (since $\mu(X) = 1$), and thus it is always σ -finite and semifinite.

Note: These same names are also applied to the restrictions of these measures to smaller σ -algebras on X .

Semifinite measure which is not σ -finite : To show that a semifinite measure is not necessarily σ -finite, consider the **counting measure** μ on an uncountable set X (for example, $X = \mathbb{R}$) with the σ -algebra $\mathcal{P}(X)$.

1. **μ is semifinite:** Let $E \subset X$ with $\mu(E) = \infty$. This means E is an infinite set. Since X consists of points, we can pick any single point $x \in E$. The set $F = \{x\}$ is a measurable subset of E , and $\mu(F) = 1$. Since $0 < 1 < \infty$, the measure is semifinite.
2. **μ is not σ -finite:** Suppose $X = \bigcup_{n=1}^{\infty} E_n$ where $\mu(E_n) < \infty$ for all n . Under the counting measure, $\mu(E_n) < \infty$ means that each E_n must be a finite set. However, a countable union of finite sets is at most countable. Since we assumed X is uncountable, no such sequence $\{E_n\}$ can exist.

Thus, μ is semifinite but not σ -finite.

More Examples

Example 5 (The Countable/Co-countable Measure). Let X be an uncountable set, and let $\mathcal{M}$ be the σ -algebra of countable or co-countable sets. Define μ on $\mathcal{M}$ by:

$$\mu(E) = \begin{cases} 0 & \text{if } E \text{ is countable} \\ 1 & \text{if } E \text{ is co-countable} \end{cases}$$

This is a valid measure because in any sequence of disjoint sets $\{E_j\} \subset \mathcal{M}$, at most one E_j can be co-countable. This ensures that countable additivity holds.

Example 6 (A Finitely Additive Function that is Not a Measure). Let X be an infinite set and $\mathcal{M} = \mathcal{P}(X)$. Define $\mu(E) = 0$ if E is finite and $\mu(E) = \infty$ if E is infinite.

- **Finite Additivity:** If A, B are disjoint and finite, $A \cup B$ is finite ($0 + 0 = 0$). If one is infinite, $A \cup B$ is infinite ($\infty + 0 = \infty$).
- **Failure of Countable Additivity:** Let $X = \mathbb{N}$ and $E_j = \{j\}$. Then $\sum \mu(E_j) = \sum 0 = 0$, but $\mu(\bigcup E_j) = \mu(\mathbb{N}) = \infty$.

Thus, μ is finitely additive but is **not** a measure.

Definition 18 (Null Set). If $(X, \mathcal{M}, \mu)$ is a measure space, a set $E \in \mathcal{M}$ such that $\mu(E) = 0$ is called a **null set**. By subadditivity, any countable union of null sets is a null set.

Definition 19 (Almost Everywhere). If a statement about points $x \in X$ is true except for x in some null set, we say that it is true **almost everywhere** (abbreviated **a.e.**), or for **almost every** x . If more precision is needed, we shall speak of a **μ -null set**, or **μ -almost everywhere**.

Definition 20 (σ -finite Set). Let $(X, \mathcal{M}, \mu)$ be a measure space. A set $E \in \mathcal{M}$ is said to be **σ -finite** for μ if $E = \bigcup_1^\infty E_j$ where $E_j \in \mathcal{M}$ and $\mu(E_j) < \infty$ for all j .

Definition 21 (Semifinite Measure). A measure μ is called **semifinite** if for each $E \in \mathcal{M}$ with $\mu(E) = \infty$, there exists some $F \in \mathcal{M}$ with $F \subset E$ and $0 < \mu(F) < \infty$.

Definition 22 (Finitely Additive Measure). A function μ that satisfies $\mu(\emptyset) = 0$ and finite additivity ($\mu(\bigcup_1^n E_j) = \sum_1^n \mu(E_j)$ for disjoint sets), but not necessarily countable additivity, is called a **finitely additive measure**.

Complete Measures

In a general measure space, if E is a **null set** ($\mu(E) = 0$) and $F \subset E$, it follows from monotonicity that $\mu(F) = 0$ provided that $F \in \mathcal{M}$. However, it is not always true that every subset of a null set belongs to the σ -algebra $\mathcal{M}$.

Definition 23. A measure whose domain includes all subsets of null sets is called **complete**.

Completeness can often simplify technical arguments and can always be achieved by enlarging the domain of the measure μ .

Theorem 3 (Completion of a Measure). Suppose that $(X, \mathcal{M}, \mu)$ is a measure space. Let $\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}$ be the collection of null sets, and define:

$$\overline{\mathcal{M}} = \{E \cup F : E \in \mathcal{M} \text{ and } F \subset N \text{ for some } N \in \mathcal{N}\}$$

Then $\overline{\mathcal{M}}$ is a σ -algebra, and there is a unique extension $\bar{\mu}$ of μ to a complete measure on $\overline{\mathcal{M}}$.

Proof. Let $(X, \mathcal{M}, \mu)$ be a measure space and $\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}$. We define:

$$\overline{\mathcal{M}} = \{E \cup F : E \in \mathcal{M} \text{ and } F \subset N \text{ for some } N \in \mathcal{N}\}$$

Step 1: Closed under Countable Unions

Let $\{A_j\}_{j=1}^\infty$ be a sequence in $\overline{\mathcal{M}}$. By definition, $A_j = E_j \cup F_j$ where $E_j \in \mathcal{M}$ and $F_j \subset N_j \in \mathcal{N}$. We look at the union $A = \bigcup A_j$:

- Let $E = \bigcup E_j$. Since $\mathcal{M}$ is a σ -algebra, $E \in \mathcal{M}$.
- Let $F = \bigcup F_j$ and $N = \bigcup N_j$. Since $\mathcal{M}$ is a σ -algebra, $N \in \mathcal{M}$.
- By subadditivity, $\mu(N) \leq \sum \mu(N_j) = 0$, so $N \in \mathcal{N}$.
- Since $F_j \subset N_j$, it follows that $F \subset N$.

Thus $A = E \cup F \in \overline{\mathcal{M}}$.

Step 2: Closed under Complements

Let $A = E \cup F \in \overline{\mathcal{M}}$ with $F \subset N \in \mathcal{N}$. We can assume $E \cap N = \emptyset$ by replacing F and N with $F \setminus E$ and $N \setminus E$ if necessary. Observe that $E \cup F = (E \cup N) \cap (N^c \cup F)$. Taking the complement:

$$(E \cup F)^c = (E \cup N)^c \cup (N \setminus F)$$

- $(E \cup N)^c \in \mathcal{M}$ because $E, N \in \mathcal{M}$.
- $N \setminus F \subset N$, and since N is a null set, $N \setminus F$ is a subset of a null set.

Thus $A^c \in \overline{\mathcal{M}}$, and $\overline{\mathcal{M}}$ is a σ -algebra.

Step 3: Well-definedness of $\bar{\mu}$

Define $\bar{\mu}(E \cup F) = \mu(E)$. Suppose $E_1 \cup F_1 = E_2 \cup F_2$ with $F_j \subset N_j \in \mathcal{N}$. Then:

$$E_1 \subset E_1 \cup F_1 = E_2 \cup F_2 \subset E_2 \cup N_2$$

By monotonicity: $\mu(E_1) \leq \mu(E_2 \cup N_2) \leq \mu(E_2) + \mu(N_2) = \mu(E_2)$. By symmetry, $\mu(E_2) \leq \mu(E_1)$. Hence $\mu(E_1) = \mu(E_2)$, so $\bar{\mu}$ is well-defined.

Step 4: Completeness of $\bar{\mu}$

Suppose $\bar{\mu}(A) = 0$ for some $A \in \bar{\mathcal{M}}$, and let $B \subset A$. $A = E \cup F$ where $\mu(E) = 0$. Since $F \subset N \in \mathcal{N}$, $A \subset E \cup N$. Let $N' = E \cup N \in \mathcal{N}$. Since $B \subset A \subset N'$, B is a subset of a null set. We can write $B = \emptyset \cup B$, so $B \in \bar{\mathcal{M}}$. Thus $\bar{\mu}$ is complete.

Step 5: Uniqueness

If ν is another measure on $\bar{\mathcal{M}}$ extending μ , then for $A = E \cup F$:

$$\mu(E) = \nu(E) \leq \nu(E \cup F) \leq \nu(E \cup N) \leq \nu(E) + \nu(N) = \mu(E) + 0$$

This forces $\nu(A) = \mu(E) = \bar{\mu}(A)$. □

The measure $\bar{\mu}$ in the above theorem is called the **completion of μ** , and $\bar{\mathcal{M}}$ is called the **completion of $\mathcal{M}$ with respect to μ** .

Outer Measures

In this section, we develop the tools used to construct measures. To motivate the ideas, it is useful to recall the procedure used in calculus to define the area of a bounded region E in the plane $\mathbb{R}^2$.

Motivation: Area in the Plane

One typically draws a grid of rectangles in the plane and approximates the area of E from two sides:

- **Inner Area:** Approximated from below by the sum of the areas of the rectangles in the grid that are *subsets* of E .
- **Outer Area:** Approximated from above by the sum of the areas of the rectangles in the grid that *intersect* E .

The limits of these approximations as the grid is taken finer and finer give the “inner area” and “outer area” of E . If they are equal, their common value is the “area” of E .

The key idea here is that of **outer area**. If R is a large rectangle containing E , the inner area of E is simply the area of R minus the outer area of $R \setminus E$. Thus, the notion of outer area is the fundamental concept we generalize to abstract spaces.

Definition of Outer Measure

The abstract generalization of the notion of outer area is as follows:

Definition 24. An **outer measure** on a nonempty set X is a function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ that satisfies the following three properties:

1. $\mu^*(\emptyset) = 0$.
2. **Monotonicity:** $\mu^*(A) \leq \mu^*(B)$ if $A \subset B$.
3. **Countable Subadditivity:** $\mu^*(\bigcup_{j=1}^{\infty} A_j) \leq \sum_{j=1}^{\infty} \mu^*(A_j)$ for any sequence $\{A_j\} \subset \mathcal{P}(X)$.

Remark. Note that the domain of an outer measure is the entire power set $\mathcal{P}(X)$, but it does not require countable additivity-only subadditivity.

Proposition (Proposition 1.10). Let $\mathcal{E} \subset \mathcal{P}(X)$ and $\rho : \mathcal{E} \rightarrow [0, \infty]$ be such that $\emptyset \in \mathcal{E}$, $X \in \mathcal{E}$, and $\rho(\emptyset) = 0$. For any $A \subset X$, define

$$\mu^*(A) = \inf \left\{ \sum_{j=1}^{\infty} \rho(E_j) : E_j \in \mathcal{E} \text{ and } A \subset \bigcup_{j=1}^{\infty} E_j \right\}$$

Then μ^* is an outer measure.

Proof. We verify the three axioms of an outer measure step-by-step.

Step 1: The definition of μ^* makes sense.

For any $A \subset X$, we must ensure that the set over which we take the infimum is non-empty. Since $X \in \mathcal{E}$ and $A \subset X$, we can choose the sequence $E_1 = X$ and $E_j = \emptyset$ for all $j > 1$. This sequence covers A , so the infimum is well-defined.

Step 2: $\mu^*(\emptyset) = 0$.

Since $\emptyset \in \mathcal{E}$ and $\rho(\emptyset) = 0$, we can cover the empty set using $E_j = \emptyset$ for all j . Then $\sum \rho(E_j) = 0$. Since μ^* is non-negative and the infimum must be ≤ 0 , we have $\mu^*(\emptyset) = 0$.

Step 3: Monotonicity.

Suppose $A \subset B$.

- Let $\mathcal{C}_A$ be the collection of all sequences $\{E_j\} \subset \mathcal{E}$ that cover A , and $\mathcal{C}_B$ be those that cover B .
- If a sequence $\{E_j\}$ covers B , it automatically covers A because $A \subset B$. Therefore, $\mathcal{C}_B \subset \mathcal{C}_A$.
- Taking the infimum over a larger set ($\mathcal{C}_A$) results in a smaller or equal value. Thus, $\inf(\mathcal{C}_A) \leq \inf(\mathcal{C}_B)$, which means $\mu^*(A) \leq \mu^*(B)$.

Step 4: Countable Subadditivity.

Suppose $\{A_j\}_{j=1}^\infty \subset \mathcal{P}(X)$ and let $\epsilon > 0$.

1. By the property of the infimum, for each j , $\mu^*(A_j) + \epsilon 2^{-j}$ is not a lower bound for the sums. Thus, there exists a sequence $\{E_j^k\}_{k=1}^\infty \subset \mathcal{E}$ such that $A_j \subset \bigcup_{k=1}^\infty E_j^k$ and:

$$\sum_{k=1}^\infty \rho(E_j^k) \leq \mu^*(A_j) + \epsilon 2^{-j}$$

2. Let $A = \bigcup_{j=1}^\infty A_j$. Then $A \subset \bigcup_{j,k=1}^\infty E_j^k$. This double union is a countable union of elements in $\mathcal{E}$, so it is a valid candidate for the infimum defining $\mu^*(A)$.
3. By the definition of μ^* as the *infimum*, we have:

$$\mu^*(A) \leq \sum_{j,k} \rho(E_j^k) = \sum_j \left(\sum_k \rho(E_j^k) \right)$$

4. Substituting the inequality from step (1):

$$\mu^*(A) \leq \sum_j (\mu^*(A_j) + \epsilon 2^{-j}) = \sum_j \mu^*(A_j) + \epsilon \sum_j 2^{-j} = \sum_j \mu^*(A_j) + \epsilon$$

Since ϵ is arbitrary, we conclude $\mu^*(A) \leq \sum_j \mu^*(A_j)$. □

Remark (Observation: Construction of Lebesgue Outer Measure). *We can apply the general result of above Proposition to $\mathbb{R}^n$ to define a measure for arbitrary sets. Let $\mathcal{E}$ be the collection of all ***n*-boxes** in $\mathbb{R}^n$, where an *n*-box is a product of *n* intervals. For any box B , let $\text{Vol}_n(B)$ denote its standard volume.*

Definition 25. *The ***n*-dimensional Lebesgue outer measure** $\mu_{L,n}^*$ of an arbitrary set $Y \subset \mathbb{R}^n$ is defined as:*

$$\mu_{L,n}^*(Y) = \inf \left\{ \sum_{j=1}^\infty \text{Vol}_n(A_j) : A_j \text{ are } n\text{-boxes with } Y \subset \bigcup_{j=1}^\infty A_j \right\}$$

By Proposition above, since $\emptyset$ is a box with volume 0 and $\mathbb{R}^n$ can be covered by boxes, $\mu_{L,n}^$ is a valid outer measure on $\mathcal{P}(\mathbb{R}^n)$. This gives us a concrete way to measure the "outer size" of any set in Euclidean space using simple geometric building blocks.*

The Carathéodory Condition

The fundamental step that leads from outer measures to measures is as follows. If μ^* is an outer measure on X , a set $A \subset X$ is called μ^* -**measurable** if it satisfies the **Carathéodory condition**:

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \text{ for all } E \subset X.$$

Key Observations for Students:

- **Subadditivity is Given:** The inequality $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c)$ holds for any A and E by the definition of an outer measure.
- **The Proof Strategy:** To prove that A is μ^* -measurable, it is sufficient to prove only the reverse inequality:

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

- **Handling Infinite Measure:** The reverse inequality is trivial if $\mu^*(E) = \infty$.
- **Simplified Criterion:** Therefore, A is μ^* -measurable if and only if:

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c) \text{ for all } E \subset X \text{ such that } \mu^*(E) < \infty.$$

Theorem 4 (Carathéodory's Theorem). *If μ^* is an outer measure on X , the collection $\mathcal{M}$ of μ^* -measurable sets is a σ -algebra, and the restriction of μ^* to $\mathcal{M}$ is a complete measure.*

Proof. Let μ^* be an outer measure on X and $\mathcal{M}$ be the collection of sets $A \subset X$ satisfying the Carathéodory condition:

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \text{ for all } E \subset X.$$

Step 1: $\mathcal{M}$ is closed under complements.

The definition of the Carathéodory condition is perfectly symmetric with respect to A and A^c . Since $(A^c)^c = A$, the equation for A^c is identical to the equation for A . Thus, $A \in \mathcal{M} \implies A^c \in \mathcal{M}$.

Step 2: $\mathcal{M}$ is an algebra and μ^* is finitely additive.

Suppose $A, B \in \mathcal{M}$. To show $A \cup B \in \mathcal{M}$, let $E \subset X$ be any test set.

1. Since $A \in \mathcal{M}$, we split E using A :

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

2. Since $B \in \mathcal{M}$, we split the pieces $(E \cap A)$ and $(E \cap A^c)$ using B :

$$\mu^*(E \cap A) = \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c)$$

$$\mu^*(E \cap A^c) = \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c)$$

3. Substitute these back into the original equation for $\mu^*(E)$:

$$\mu^*(E) = \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c)$$

4. Note that $A \cup B = (A \cap B) \cup (A \cap B^c) \cup (A^c \cap B)$. By the subadditivity of μ^* :

$$\mu^*(E \cap (A \cup B)) \leq \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B)$$

5. Since $A^c \cap B^c = (A \cup B)^c$, substituting the inequality from (4) into (3) yields:

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$$

This proves $A \cup B \in \mathcal{M}$. If A, B are disjoint, setting $E = A \cup B$ in the split $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$ gives $\mu^*(A \cup B) = \mu^*(A) + \mu^*(B)$, showing finite additivity.

Step 3: $\mathcal{M}$ is a σ -algebra.

Let $\{A_j\}_1^\infty$ be disjoint sets in $\mathcal{M}$. Let $B_n = \bigcup_1^n A_j$ and $B = \bigcup_1^\infty A_j$.

1. By finite additivity and induction: $\mu^*(E \cap B_n) = \sum_{j=1}^n \mu^*(E \cap A_j)$.
2. Since $B_n \in \mathcal{M}$, we have $\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c)$.
3. By monotonicity, $B^c \subset B_n^c$, so $\mu^*(E \cap B_n^c) \geq \mu^*(E \cap B^c)$. Thus:

$$\mu^*(E) \geq \sum_{j=1}^n \mu^*(E \cap A_j) + \mu^*(E \cap B^c)$$

4. Taking $n \rightarrow \infty$: $\mu^*(E) \geq \sum_{j=1}^{\infty} \mu^*(E \cap A_j) + \mu^*(E \cap B^c)$.
5. By countable subadditivity: $\sum_{j=1}^{\infty} \mu^*(E \cap A_j) \geq \mu^*(\bigcup (E \cap A_j)) = \mu^*(E \cap B)$.
6. Therefore: $\mu^*(E) \geq \mu^*(E \cap B) + \mu^*(E \cap B^c)$, which proves $B \in \mathcal{M}$.

Step 4: μ^* is countably additive on $\mathcal{M}$.

Setting $E = B$ in the inequality $\mu^*(E) \geq \sum \mu^*(E \cap A_j) + \mu^*(E \cap B^c)$ gives:

$$\mu^*(B) \geq \sum_{j=1}^{\infty} \mu^*(A_j) + 0$$

Since $\mu^*(B) \leq \sum \mu^*(A_j)$ by subadditivity, we have equality.

Step 5: The measure is complete.

If $\mu^*(A) = 0$, then for any $E \subset X$, $\mu^*(E \cap A) = 0$ by monotonicity. Since $\mu^*(E) \geq \mu^*(E \cap A^c)$ also by monotonicity, we have:

$$\mu^*(E) \geq 0 + \mu^*(E \cap A^c) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

This satisfies the Carathéodory condition, so $A \in \mathcal{M}$. □

1 Integration

Example 7. We recall that any mapping $f : X \rightarrow Y$ between two sets induces a mapping $f^{-1} : \mathcal{P}(Y) \rightarrow \mathcal{P}(X)$, defined by $f^{-1}(E) = \{x \in X : f(x) \in E\}$, which preserves unions, intersections, and complements. Thus, if $\mathcal{N}$ is a σ -algebra on Y , the collection $\{f^{-1}(E) : E \in \mathcal{N}\}$ is a σ -algebra on X .

Verification:

- (i) **Empty set:** Since $\mathcal{N}$ is a σ -algebra, $\emptyset \in \mathcal{N}$. Thus, $f^{-1}(\emptyset) = \emptyset$ belongs to the collection.
- (ii) **Complements:** If $f^{-1}(E)$ is in the collection for some $E \in \mathcal{N}$, then its complement is given by $(f^{-1}(E))^c = f^{-1}(E^c)$. Since $\mathcal{N}$ is closed under complements, $E^c \in \mathcal{N}$, so $f^{-1}(E^c)$ belongs to the collection.
- (iii) **Countable unions:** If $\{f^{-1}(E_j)\}_{j=1}^{\infty}$ is a sequence in the collection with $E_j \in \mathcal{N}$, then $\bigcup_{j=1}^{\infty} f^{-1}(E_j) = f^{-1}\left(\bigcup_{j=1}^{\infty} E_j\right)$. Since $\mathcal{N}$ is closed under countable unions, $\bigcup_{j=1}^{\infty} E_j \in \mathcal{N}$, so the union belongs to the collection.

Definition 26. If $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measurable spaces, a mapping $f : X \rightarrow Y$ is called $(\mathcal{M}, \mathcal{N})$ -measurable, or just **measurable** when $\mathcal{M}$ and $\mathcal{N}$ are understood, if $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{N}$.

This proposition provides a highly practical shortcut for checking measurability. Instead of verifying that the preimage of *every* set in the σ -algebra $\mathcal{N}$ is measurable, we only need to check the preimages of the smaller collection of sets that generate $\mathcal{N}$.

Proposition. If $\mathcal{N}$ is generated by $\mathcal{E}$, then $f : X \rightarrow Y$ is $(\mathcal{M}, \mathcal{N})$ -measurable iff $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{E}$.

Proof. Forward Direction ($\implies$):

Assume f is $(\mathcal{M}, \mathcal{N})$ -measurable. By definition, $f^{-1}(E) \in \mathcal{M}$ for every $E \in \mathcal{N}$. Since $\mathcal{E} \subset \mathcal{N}$, it follows immediately that $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{E}$.

Backward Direction ($\impliedby$):

Assume that $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{E}$. Let us define the collection:

$$\mathcal{A} = \{E \subset Y : f^{-1}(E) \in \mathcal{M}\}$$

Our goal is to show that $\mathcal{N} \subset \mathcal{A}$. We proceed by verifying two facts:

(i) **$\mathcal{A}$ contains the generator:** By our initial assumption, $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{E}$, which means $\mathcal{E} \subset \mathcal{A}$.

(ii) **$\mathcal{A}$ is a σ -algebra on Y :**

- $f^{-1}(\emptyset) = \emptyset \in \mathcal{M} \implies \emptyset \in \mathcal{A}$.
- If $E \in \mathcal{A}$, then $f^{-1}(E) \in \mathcal{M}$. Since $\mathcal{M}$ is a σ -algebra, $(f^{-1}(E))^c = f^{-1}(E^c) \in \mathcal{M}$, which implies $E^c \in \mathcal{A}$.
- If $\{E_j\}_{j=1}^\infty \subset \mathcal{A}$, then $f^{-1}(E_j) \in \mathcal{M}$ for each j . Since $\mathcal{M}$ is closed under countable unions, $\bigcup_{j=1}^\infty f^{-1}(E_j) = f^{-1}\left(\bigcup_{j=1}^\infty E_j\right) \in \mathcal{M}$, which implies $\bigcup_{j=1}^\infty E_j \in \mathcal{A}$.

Since $\mathcal{A}$ is a σ -algebra containing $\mathcal{E}$, and $\mathcal{N}$ is the *smallest* σ -algebra containing $\mathcal{E}$ (as $\mathcal{N} = \sigma(\mathcal{E})$), it must be that $\mathcal{N} \subset \mathcal{A}$.

Therefore, for every $E \in \mathcal{N}$, we have $E \in \mathcal{A}$, meaning $f^{-1}(E) \in \mathcal{M}$. This proves that f is $(\mathcal{M}, \mathcal{N})$ -measurable. $\square$

Corollary 5. *If X and Y are metric (or topological) spaces, every continuous $f : X \rightarrow Y$ is $(\mathcal{B}_X, \mathcal{B}_Y)$ -measurable.*

Proof. Recall that by the topological definition of continuity, a mapping $f : X \rightarrow Y$ is continuous if and only if $f^{-1}(U)$ is open in X for every open set $U \subset Y$.

Let $\mathcal{E}$ be the collection of all open sets in Y . By definition, the Borel σ -algebra $\mathcal{B}_Y$ is generated by $\mathcal{E}$ (i.e., $\mathcal{B}_Y = \sigma(\mathcal{E})$). Since f is continuous, for every open set $U \in \mathcal{E}$, its preimage $f^{-1}(U)$ is open in X , and therefore $f^{-1}(U) \in \mathcal{B}_X$.

Applying the previous Proposition, since the preimages of all generating sets $U \in \mathcal{E}$ are measurable in X , it follows that f is $(\mathcal{B}_X, \mathcal{B}_Y)$ -measurable. $\square$

Example 8. *The composition of measurable mappings is measurable; that is, if $f : X \rightarrow Y$ is $(\mathcal{M}, \mathcal{N})$ -measurable and $g : Y \rightarrow Z$ is $(\mathcal{N}, \mathcal{O})$ -measurable, then $g \circ f : X \rightarrow Z$ is $(\mathcal{M}, \mathcal{O})$ -measurable.*

Proof: To show that $g \circ f$ is $(\mathcal{M}, \mathcal{O})$ -measurable, we must verify that $(g \circ f)^{-1}(E) \in \mathcal{M}$ for any measurable set $E \in \mathcal{O}$.

Let $E \in \mathcal{O}$. Using the properties of preimages for composite functions, we have:

$$(g \circ f)^{-1}(E) = f^{-1}(g^{-1}(E))$$

Since $g : Y \rightarrow Z$ is $(\mathcal{N}, \mathcal{O})$ -measurable, the preimage $g^{-1}(E)$ belongs to $\mathcal{N}$. Furthermore, since $f : X \rightarrow Y$ is $(\mathcal{M}, \mathcal{N})$ -measurable, the preimage under f of any set in $\mathcal{N}$ belongs to $\mathcal{M}$. Therefore, $f^{-1}(g^{-1}(E)) \in \mathcal{M}$, which proves that $g \circ f$ is $(\mathcal{M}, \mathcal{O})$ -measurable.

Remark. If $(X, \mathcal{M})$ is a measurable space, a real- or complex-valued function f on X will be called **$\mathcal{M}$ -measurable**, or just **measurable**, if it is $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$ or $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$ measurable. $\mathcal{B}_{\mathbb{R}}$ or $\mathcal{B}_{\mathbb{C}}$ is always understood as the σ -algebra on the range space unless otherwise specified.

In particular, $f : \mathbb{R} \rightarrow \mathbb{C}$ is **Lebesgue** (resp. **Borel**) **measurable** if it is $(\mathcal{L}, \mathcal{B}_{\mathbb{C}})$ (resp. $(\mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{C}})$) measurable; likewise for $f : \mathbb{R} \rightarrow \mathbb{R}$.

Proposition. *If $(X, \mathcal{M})$ is a measurable space and $f : X \rightarrow \mathbb{R}$, the following are equivalent:*

- a. f is $\mathcal{M}$ -measurable.
- b. $f^{-1}((a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$.

- c. $f^{-1}([a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$.
- d. $f^{-1}((-\infty, a)) \in \mathcal{M}$ for all $a \in \mathbb{R}$.
- e. $f^{-1}((-\infty, a]) \in \mathcal{M}$ for all $a \in \mathbb{R}$.

Proof. We will prove the equivalence by showing the implications $a \implies b \implies c \implies e \implies d \implies b$ and finally showing how any of these implies a.

Step 1: (a) $\implies$ (b)

Assume f is $\mathcal{M}$ -measurable. By definition, f is $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$ -measurable, meaning $f^{-1}(E) \in \mathcal{M}$ for every Borel set $E \in \mathcal{B}_{\mathbb{R}}$. Since the open interval (a, ∞) is an open set in $\mathbb{R}$, it is a Borel set. Thus, $f^{-1}((a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$.

Step 2: (b) $\implies$ (c)

Assume $f^{-1}((a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$. We can express the closed-open interval $[a, \infty)$ as a countable intersection of open intervals:

$$[a, \infty) = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, \infty \right)$$

Taking the preimage on both sides gives:

$$f^{-1}([a, \infty)) = f^{-1} \left(\bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, \infty \right) \right) = \bigcap_{n=1}^{\infty} f^{-1} \left(\left(a - \frac{1}{n}, \infty \right) \right)$$

By assumption, each $f^{-1}((a - 1/n, \infty)) \in \mathcal{M}$. Since $\mathcal{M}$ is a σ -algebra, it is closed under countable intersections. Therefore, $f^{-1}([a, \infty)) \in \mathcal{M}$.

Step 3: (c) $\implies$ (e)

Assume $f^{-1}([a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$. The interval $(-\infty, a]$ is the complement of the interval (a, ∞) , but it can also be linked directly via complements to $[a, \infty)$ by noting that $(-\infty, a] = \mathbb{R} \setminus (a, \infty)$. Alternatively, using standard complements:

$$(-\infty, a] = \mathbb{R} \setminus [a, \infty)$$

Let's establish (c) $\implies$ (d) first for elegance:

$$f^{-1}((-\infty, a)) = f^{-1}(\mathbb{R} \setminus [a, \infty)) = (f^{-1}([a, \infty)))^c$$

Since $f^{-1}([a, \infty)) \in \mathcal{M}$ and $\mathcal{M}$ is closed under complements, $f^{-1}((-\infty, a)) \in \mathcal{M}$, which proves (d).

To get (e) from here, observe that:

$$(-\infty, a] = \bigcap_{n=1}^{\infty} \left(-\infty, a + \frac{1}{n} \right)$$

Taking preimages:

$$f^{-1}((-\infty, a]) = \bigcap_{n=1}^{\infty} f^{-1} \left(\left(-\infty, a + \frac{1}{n} \right) \right)$$

Since each term in the intersection belongs to $\mathcal{M}$ by statement (d), their countable intersection belongs to $\mathcal{M}$. Thus, $f^{-1}((-\infty, a]) \in \mathcal{M}$.

Step 4: (e) $\implies$ (d)

Assume $f^{-1}((-\infty, a]) \in \mathcal{M}$ for all $a \in \mathbb{R}$. We can write the open interval $(-\infty, a)$ as a countable union of closed-bounded-above intervals:

$$(-\infty, a) = \bigcup_{n=1}^{\infty} \left(-\infty, a - \frac{1}{n} \right]$$

Taking preimages:

$$f^{-1}((-\infty, a)) = \bigcup_{n=1}^{\infty} f^{-1} \left(\left(-\infty, a - \frac{1}{n} \right] \right)$$

Since $\mathcal{M}$ is closed under countable unions and each component belongs to $\mathcal{M}$ by statement (e), $f^{-1}((-\infty, a)) \in \mathcal{M}$.

Step 5: (d) $\implies$ (b)

Assume $f^{-1}((-\infty, a)) \in \mathcal{M}$ for all $a \in \mathbb{R}$. Since (a, ∞) is the complement of $(-\infty, a]$, we write:

$$(a, \infty) = \mathbb{R} \setminus (-\infty, a]$$

From Step 4, we know statement (e) holds if (d) holds because they are duals. Thus, taking the complement:

$$f^{-1}((a, \infty)) = (f^{-1}((-\infty, a]))^c$$

Since $f^{-1}((-\infty, a]) \in \mathcal{M}$, its complement $f^{-1}((a, \infty)) \in \mathcal{M}$. Thus, statements (b), (c), (d), and (e) are completely equivalent to one another.

Step 6: Equivalence with (a)

To complete the proof, we must show that any of these conditions implies (a). Let us take condition (b).

Assume $f^{-1}((a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$. Let $\mathcal{E} = \{(a, \infty) : a \in \mathbb{R}\}$. It is a standard result in real analysis that the collection of open rays $\mathcal{E}$ generates the Borel σ -algebra on $\mathbb{R}$ (i.e., $\sigma(\mathcal{E}) = \mathcal{B}_{\mathbb{R}}$).

By our earlier Proposition, if the preimage of every generating set in $\mathcal{E}$ belongs to $\mathcal{M}$, then the mapping f is $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$ -measurable. Therefore, f is $\mathcal{M}$ -measurable. $\square$

Definition 27. If $(X, \mathcal{M})$ is a measurable space, f is a function on X , and $E \in \mathcal{M}$, we say that f is **measurable on E** if:

$$f^{-1}(B) \cap E \in \mathcal{M} \quad \text{for all Borel sets } B.$$

Equivalently, the restriction $f|_E$ is $\mathcal{M}_E$ -measurable, where $\mathcal{M}_E = \{F \cap E : F \in \mathcal{M}\}$ is the subspace σ -algebra on E .

Remark. Explanation: This definition allows us to discuss the measurability of a function restricted to a specific domain patch E without requiring f to be well-behaved or even measurable across the entire space X . Notice that $f^{-1}(B) \cap E$ is precisely $(f|_E)^{-1}(B)$, making it a natural internalization of subspace measurability.

Definition 28. Given a set X and a family of measurable spaces $\{(Y_\alpha, \mathcal{N}_\alpha)\}_{\alpha \in A}$ with a map $f_\alpha : X \rightarrow Y_\alpha$ for each $\alpha \in A$, the **σ -algebra generated by $\{f_\alpha\}_{\alpha \in A}$** is the unique smallest σ -algebra on X with respect to which all the f_α 's are measurable. It is explicitly generated by the collection of sets:

$$\{f_\alpha^{-1}(E_\alpha) : E_\alpha \in \mathcal{N}_\alpha \text{ and } \alpha \in A\}$$

Definition 29. In particular, if $X = \prod_{\alpha \in A} Y_\alpha$ is a cartesian product space, the **product σ -algebra** on X is defined as the σ -algebra generated by the coordinate projection maps $\pi_\alpha : X \rightarrow Y_\alpha$.

Remark. Explanation: Think of this as pulling back information. If X doesn't have an inherent structure yet, we can build a tailor-made σ -algebra on it by taking all necessary preimages from the target spaces Y_α so that every function f_α is forced to satisfy the definition of measurability. In the case of product spaces, this construction guarantees that working with a multi-dimensional structure scales down predictably to checking properties on individual target component paths via coordinates π_α .

Proposition. Let $(X, \mathcal{M})$ and $(Y_\alpha, \mathcal{N}_\alpha)$ ($\alpha \in A$) be measurable spaces, $Y = \prod_{\alpha \in A} Y_\alpha$, $\mathcal{N} = \bigotimes_{\alpha \in A} \mathcal{N}_\alpha$, and $\pi_\alpha : Y \rightarrow Y_\alpha$ the coordinate projection maps. Then a mapping $f : X \rightarrow Y$ is $(\mathcal{M}, \mathcal{N})$ -measurable iff $f_\alpha = \pi_\alpha \circ f$ is $(\mathcal{M}, \mathcal{N}_\alpha)$ -measurable for all $\alpha \in A$.

Proof. Forward Direction ($\implies$):

Assume $f : X \rightarrow Y$ is $(\mathcal{M}, \mathcal{N})$ -measurable. By definition, the coordinate projection map $\pi_\alpha : Y \rightarrow Y_\alpha$ is $(\mathcal{N}, \mathcal{N}_\alpha)$ -measurable because the product σ -algebra $\mathcal{N}$ is constructed precisely to make these projection maps measurable.

Since $f_\alpha = \pi_\alpha \circ f$ is the composition of two measurable mappings (f and π_α), it follows naturally that f_α is $(\mathcal{M}, \mathcal{N}_\alpha)$ -measurable for each $\alpha \in A$.

Backward Direction ($\impliedby$):

Assume that each component function $f_\alpha = \pi_\alpha \circ f$ is $(\mathcal{M}, \mathcal{N}_\alpha)$ -measurable for all $\alpha \in A$.

To show that f is $(\mathcal{M}, \mathcal{N})$ -measurable, we must prove that $f^{-1}(B) \in \mathcal{M}$ for every set $B \in \mathcal{N}$. One of our previous propositions states that instead of checking the inverse image of every single measurable

set, it suffices to prove this condition just for the collection of sets that generates the σ -algebra. We will perform that verification directly here.

Let us define the collection of all subsets in Y whose preimages under f are measurable in X :

$$\mathcal{A} = \{B \subset Y : f^{-1}(B) \in \mathcal{M}\}$$

We first check that $\mathcal{A}$ forms a valid σ -algebra on Y :

- $f^{-1}(\emptyset) = \emptyset \in \mathcal{M} \implies \emptyset \in \mathcal{A}$.
- If $B \in \mathcal{A}$, then $f^{-1}(B) \in \mathcal{M}$. Since $\mathcal{M}$ is a σ -algebra, its complement $(f^{-1}(B))^c = f^{-1}(B^c) \in \mathcal{M}$, which implies $B^c \in \mathcal{A}$.
- If $\{B_j\}_{j=1}^\infty \subset \mathcal{A}$, then $f^{-1}(B_j) \in \mathcal{M}$ for each j . Since $\mathcal{M}$ is closed under countable unions, $\bigcup_{j=1}^\infty f^{-1}(B_j) = f^{-1}\left(\bigcup_{j=1}^\infty B_j\right) \in \mathcal{M}$, which implies $\bigcup_{j=1}^\infty B_j \in \mathcal{A}$.

Thus, $\mathcal{A}$ is a σ -algebra on Y .

Now, recall that the product σ -algebra $\mathcal{N}$ is defined as the smallest σ -algebra generated by the elementary cylinder sets of the form $\pi_\alpha^{-1}(E_\alpha)$ where $E_\alpha \in \mathcal{N}_\alpha$ and $\alpha \in A$. Let us test if these generating sets belong to $\mathcal{A}$. Taking the preimage under f :

$$f^{-1}(\pi_\alpha^{-1}(E_\alpha)) = (\pi_\alpha \circ f)^{-1}(E_\alpha) = f_\alpha^{-1}(E_\alpha)$$

By our initial assumption, each component function f_α is measurable, meaning $f_\alpha^{-1}(E_\alpha) \in \mathcal{M}$. Therefore, every generating set $\pi_\alpha^{-1}(E_\alpha)$ belongs to $\mathcal{A}$.

Since $\mathcal{A}$ is a σ -algebra containing all the generating sets of Y , and $\mathcal{N}$ is the unique smallest σ -algebra containing those same generating sets, it must hold that $\mathcal{N} \subset \mathcal{A}$. Consequently, for every $B \in \mathcal{N}$, we have $B \in \mathcal{A}$, meaning $f^{-1}(B) \in \mathcal{M}$. This proves that f is $(\mathcal{M}, \mathcal{N})$ -measurable. $\square$

Corollary 6. *A function $f : X \rightarrow \mathbb{C}$ is $\mathcal{M}$ -measurable iff $\operatorname{Re} f$ and $\operatorname{Im} f$ are $\mathcal{M}$ -measurable.*

Proof. We can identify the complex plane $\mathbb{C}$ topologically with $\mathbb{R}^2$ via the standard mapping $z \mapsto (\operatorname{Re} z, \operatorname{Im} z)$. Under this identification, the Borel σ -algebra on $\mathbb{C}$ satisfies:

$$\mathcal{B}_{\mathbb{C}} = \mathcal{B}_{\mathbb{R}^2} = \mathcal{B}_{\mathbb{R}} \otimes \mathcal{B}_{\mathbb{R}}$$

where $\mathcal{B}_{\mathbb{R}} \otimes \mathcal{B}_{\mathbb{R}}$ denotes the product σ -algebra on $\mathbb{R} \times \mathbb{R}$.

Consider the coordinate projection maps $\pi_1 : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $\pi_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $\pi_1(x, y) = x$ and $\pi_2(x, y) = y$. Under our identification, these projections correspond exactly to taking the real and imaginary parts of a complex number:

$$\pi_1 \circ f = \operatorname{Re} f \quad \text{and} \quad \pi_2 \circ f = \operatorname{Im} f$$

By the above Proposition, a mapping into a product space equipped with its product σ -algebra is measurable if and only if its composition with each coordinate projection map is measurable. Applying this directly to $f : X \rightarrow \mathbb{R}^2$, it follows that f is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$ -measurable if and only if $\pi_1 \circ f = \operatorname{Re} f$ and $\pi_2 \circ f = \operatorname{Im} f$ are both $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$ -measurable. $\square$

Remark (Functions with Values in the Extended Real Number System). *In measure theory and integration, we frequently encounter functions whose values can become infinitely large (such as the limit of an increasing sequence of positive functions, or the measure of an unbounded set). To handle these situations rigorously without breaking our mathematical framework, we extend the real line by adding positive and negative infinity.*

1. Defining the Extended Real Number System ($\overline{\mathbb{R}}$)

The **extended real number system**, denoted by $\overline{\mathbb{R}}$, is defined as the standard real line $\mathbb{R}$ unioned with the two symbolic endpoints $\{-\infty, +\infty\}$:

$$\overline{\mathbb{R}} = [-\infty, \infty] = \mathbb{R} \cup \{-\infty, +\infty\}$$

2. The Extended Borel σ -Algebra ($\mathcal{B}_{\overline{\mathbb{R}}}$)

To define measurability for functions mapping into $\overline{\mathbb{R}}$, we must establish what it means for a subset of $\overline{\mathbb{R}}$ to be a Borel set. We define the extended Borel σ -algebra, $\mathcal{B}_{\overline{\mathbb{R}}}$, by tracking how sets intersect with the standard real numbers:

$$\mathcal{B}_{\overline{\mathbb{R}}} = \{E \subset \overline{\mathbb{R}} : E \cap \mathbb{R} \in \mathcal{B}_{\mathbb{R}}\}$$

What this means for students: A set E is an extended Borel set if its "finite part" ($E \cap \mathbb{R}$) is a standard real Borel set. The set E itself may or may not contain the isolated points $\{+\infty\}$ and $\{-\infty\}$.

3. Topological View: The Arc-Tangent Distance

An alternative and elegant way to visualize $\overline{\mathbb{R}}$ is to treat it as a legitimate metric space. We can "squish" the infinite line into a finite interval using the homeomorphism $A(x) = \arctan x$.

- The function $\arctan x$ maps the infinite line $\mathbb{R}$ bijectively onto the open interval $(-\pi/2, \pi/2)$.
- By assigning $\arctan(\infty) = \pi/2$ and $\arctan(-\infty) = -\pi/2$, the extended system $\overline{\mathbb{R}}$ maps perfectly onto the closed, bounded interval $[-\pi/2, \pi/2]$.

We can then define a rigorous metric (distance function) ρ on $\overline{\mathbb{R}}$ by:

$$\rho(x, y) = |\arctan x - \arctan y|$$

The open sets generated by this metric give rise to a topology whose Borel sets are exactly the collection $\mathcal{B}_{\overline{\mathbb{R}}}$ defined above.

4. Checking Measurability via One-Sided Rays

By applying our earlier proposition (Proposition 2.1), we do not need to check the preimage of every single complex set in $\mathcal{B}_{\overline{\mathbb{R}}}$ to prove a function is measurable. The σ -algebra $\mathcal{B}_{\overline{\mathbb{R}}}$ can be completely generated by simple, one-sided extended intervals (rays).

Therefore, a function $f : X \rightarrow \overline{\mathbb{R}}$ is defined to be $\mathcal{M}$ -**measurable** if it is $(\mathcal{M}, \mathcal{B}_{\overline{\mathbb{R}}})$ -measurable, which is equivalent to verifying just one of the following straightforward conditions for all $a \in \mathbb{R}$:

- (i) $f^{-1}((a, \infty]) \in \mathcal{M}$ (the preimage of every ray extending up to positive infinity is measurable)
- (ii) $f^{-1}([-\infty, a)) \in \mathcal{M}$ (the preimage of every ray extending down to negative infinity is measurable)

Proposition. If $f, g : X \rightarrow \mathbb{C}$ are $\mathcal{M}$ -measurable, then so are $f + g$ and fg .

Proof. To make this proof perfectly intuitive for students, we break it down into three logical phases: constructing a joint mapping into a product space, verifying the continuity of arithmetic operations, and analyzing their compositions.

Phase 1: Defining a Joint Function into a Product Space

Let us combine our two individual functions f and g into a single, joint mapping $F : X \rightarrow \mathbb{C} \times \mathbb{C}$ defined by:

$$F(x) = (f(x), g(x))$$

We need to determine if F is a measurable function. By identifying $\mathbb{C}$ with $\mathbb{R}^2$, the Borel σ -algebra on the product space satisfies $\mathcal{B}_{\mathbb{C} \times \mathbb{C}} = \mathcal{B}_{\mathbb{C}} \otimes \mathcal{B}_{\mathbb{C}}$.

Recall from Proposition 2.4 that a multi-dimensional function mapping into a product space is measurable if and only if each of its component projection coordinate paths is individually measurable. The coordinate projections of F are exactly our starting functions:

$$\pi_1 \circ F = f \quad \text{and} \quad \pi_2 \circ F = g$$

Since we are given that f and g are $\mathcal{M}$ -measurable, both component projections are measurable. Therefore, by Proposition 2.4, the joint map F is $(\mathcal{M}, \mathcal{B}_{\mathbb{C} \times \mathbb{C}})$ -measurable.

Phase 2: The Measurability of Arithmetic Operations

Now let us look at the target operations themselves. Define the addition map $\phi : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ and the multiplication map $\psi : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ as follows:

$$\phi(z, w) = z + w \quad \text{and} \quad \psi(z, w) = zw$$

From basic topology, we know that standard algebraic operations—addition and multiplication of complex numbers—are continuous functions.

By Corollary 2.2, any continuous function between topological spaces is automatically measurable with respect to their corresponding Borel σ -algebras. Therefore, ϕ and ψ are $(\mathcal{B}_{\mathbb{C} \times \mathbb{C}}, \mathcal{B}_{\mathbb{C}})$ -measurable mappings.

Phase 3: Composition and Conclusion

Finally, notice that the functions $f + g$ and fg can be written perfectly as compositions of our operational maps with our joint map:

$$(f + g)(x) = f(x) + g(x) = \phi(f(x), g(x)) = (\phi \circ F)(x)$$

$$(fg)(x) = f(x)g(x) = \psi(f(x), g(x)) = (\psi \circ F)(x)$$

We have already established that:

- (i) F is a measurable mapping from X to $\mathbb{C} \times \mathbb{C}$.
- (ii) ϕ and ψ are measurable mappings from $\mathbb{C} \times \mathbb{C}$ to $\mathbb{C}$.

Since the composition of two measurable mappings is always measurable, it follows immediately that $f + g = \phi \circ F$ and $fg = \psi \circ F$ are $\mathcal{M}$ -measurable functions. $\square$

Proposition. *If $\{f_j\}$ is a sequence of $\overline{\mathbb{R}}$ -valued measurable functions on $(X, \mathcal{M})$, then the functions*

$$\begin{aligned} g_1(x) &= \sup_j f_j(x), & g_3(x) &= \limsup_{j \rightarrow \infty} f_j(x), \\ g_2(x) &= \inf_j f_j(x), & g_4(x) &= \liminf_{j \rightarrow \infty} f_j(x) \end{aligned}$$

are all measurable. If $f(x) = \lim_{j \rightarrow \infty} f_j(x)$ exists for every $x \in X$, then f is measurable.

Proof. To make this proof accessible to students, we will break it down into clear stages. We leverage the fact that for extended real-valued functions ($\overline{\mathbb{R}}$), we can establish full measurability by checking only one-sided infinite rays like $(a, \infty]$ or $[-\infty, a]$.

Step 1: Proving the Measurability of the Supremum (g_1)

Let $g_1(x) = \sup_j f_j(x)$. We choose to evaluate the preimage of the open ray $(a, \infty]$ for an arbitrary $a \in \mathbb{R}$.

By definition of the supremum, the value $g_1(x) = \sup_j f_j(x)$ is strictly greater than a if and only if ****at least one**** individual function $f_j(x)$ in the sequence exceeds a . Translating this logical "at least one" statement into set theory gives us a countable union of preimages:

$$g_1^{-1}((a, \infty]) = \bigcup_{j=1}^{\infty} f_j^{-1}((a, \infty])$$

Since every function f_j in the sequence is given to be $\mathcal{M}$ -measurable, each individual set $f_j^{-1}((a, \infty])$ belongs to the σ -algebra $\mathcal{M}$. Because a σ -algebra is closed under countable unions, their union must also belong to $\mathcal{M}$. Therefore, $g_1 = \sup_j f_j$ is $\mathcal{M}$ -measurable.

Step 2: Proving the Measurability of the Infimum (g_2)

Let $g_2(x) = \inf_j f_j(x)$. This time, it is contextually simpler to check the downward ray $[-\infty, a)$ for an arbitrary $a \in \mathbb{R}$.

By definition of the infimum, the greatest lower bound $g_2(x) = \inf_j f_j(x)$ is strictly less than a if and only if ****at least one**** individual function $f_j(x)$ drops strictly below a . Once again, this logical "at least one" corresponds to a countable union:

$$g_2^{-1}([-\infty, a)) = \bigcup_{j=1}^{\infty} f_j^{-1}([-\infty, a))$$

Because each f_j is $\mathcal{M}$ -measurable, each set $f_j^{-1}([-\infty, a)) \in \mathcal{M}$. Since $\mathcal{M}$ is closed under countable unions, the resulting union belongs to $\mathcal{M}$. This proves $g_2 = \inf_j f_j$ is $\mathcal{M}$ -measurable.

Step 3: Proving the Measurability of the Limit Superior (g_3)

Recall the standard analytical definition of the limit superior as a sequential optimization:

$$\limsup_{j \rightarrow \infty} f_j(x) = \inf_{k \geq 1} \left(\sup_{j > k} f_j(x) \right)$$

Let us analyze this from the inside out:

1. For a fixed integer k , define the "tail supremum" function:

$$h_k(x) = \sup_{j > k} f_j(x)$$

Since h_k is the supremum of a countable sub-sequence of our measurable functions, it is measurable by the exact logic we established in **Step 1**.

2. Now, notice that the total limit superior function $g_3(x)$ can be rewritten as:

$$g_3(x) = \inf_{k \geq 1} h_k(x)$$

Since g_3 is the infimum of a countable collection of functions $\{h_k\}_{k=1}^{\infty}$ which we just proved are measurable, it must be measurable by the exact logic established in **Step 2**.

Thus, $g_3 = \limsup f_j$ is $\mathcal{M}$ -measurable.

Step 4: Proving the Measurability of the Limit Inferior (g_4)

Similarly, the limit inferior is defined as:

$$\liminf_{j \rightarrow \infty} f_j(x) = \sup_{k \geq 1} \left(\inf_{j > k} f_j(x) \right)$$

Applying the identical two-step composition logic:

1. For a fixed k , the tail infimum $m_k(x) = \inf_{j > k} f_j(x)$ is a countable infimum of measurable functions, making it measurable per **Step 2**.
2. The final operation, $g_4(x) = \sup_{k \geq 1} m_k(x)$, represents a countable supremum of these measurable tail functions, making it fully measurable per **Step 1**.

Thus, $g_4 = \liminf f_j$ is $\mathcal{M}$ -measurable.

Step 5: Analysis of the Pointwise Limit (f)

Finally, suppose the pointwise limit $f(x) = \lim_{j \rightarrow \infty} f_j(x)$ exists for every $x \in X$.

From real analysis, if a sequence of numbers converges, its limit superior and its limit inferior must be exactly equal to each other, and equal to the limit itself:

$$f(x) = \limsup_{j \rightarrow \infty} f_j(x) = \liminf_{j \rightarrow \infty} f_j(x)$$

Because we have already proven that $g_3 = \limsup f_j$ is a perfectly valid $\mathcal{M}$ -measurable function, and $f(x) = g_3(x)$ everywhere, it follows immediately that f is $\mathcal{M}$ -measurable. $\square$

Corollary 7 (Corollary 2.8). *If $f, g : X \rightarrow \overline{\mathbb{R}}$ are $\mathcal{M}$ -measurable, then so are $\max(f, g)$ and $\min(f, g)$.*

Proof. To make this intuitive for students, we can connect the maximum and minimum of two functions directly to the supremum and infimum operations we established in Proposition 2.7.

Consider the maximum function, $h_1(x) = \max(f(x), g(x))$. For any given point $x \in X$, the maximum of these two values is equivalent to taking the supremum over a finite sequence containing just these two functions:

$$\max(f(x), g(x)) = \sup_{j \in \{1, 2\}} \phi_j(x) \quad \text{where } \phi_1 = f \text{ and } \phi_2 = g$$

According to Proposition 2.7, the supremum of any countable family of extended real-valued measurable functions is automatically measurable. Since a finite set is a trivial case of a countable set, $\max(f, g)$ is $\mathcal{M}$ -measurable.

Similarly, consider the minimum function, $h_2(x) = \min(f(x), g(x))$. We can reframe this as an infimum operation over the same pair of functions:

$$\min(f(x), g(x)) = \inf_{j \in \{1, 2\}} \phi_j(x)$$

By the second part of Proposition 2.7, the infimum of a countable family of measurable functions is measurable. Therefore, $\min(f, g)$ is $\mathcal{M}$ -measurable. $\square$

Corollary 8. *If $\{f_j\}$ is a sequence of complex-valued measurable functions and $f(x) = \lim_{j \rightarrow \infty} f_j(x)$ exists for all $x \in X$, then f is $\mathcal{M}$ -measurable.*

Proof. This proof reduces the complex convergence problem down to separate real convergence problems by leveraging real and imaginary parts.

Let each function f_j in the sequence be decomposed into its real component functions:

$$f_j(x) = u_j(x) + iv_j(x) \quad \text{where } u_j = \operatorname{Re} f_j \text{ and } v_j = \operatorname{Im} f_j$$

By Corollary 2.5, since each f_j is given to be complex-measurable, its real part sequence $\{u_j\}_{j=1}^{\infty}$ and its imaginary part sequence $\{v_j\}_{j=1}^{\infty}$ are sequences of real-valued $\mathcal{M}$ -measurable functions.

We are told that the pointwise limit function $f(x)$ exists everywhere. Let us decompose this limit function as well:

$$f(x) = u(x) + iv(x) \quad \text{where } u = \operatorname{Re} f \text{ and } v = \operatorname{Im} f$$

From the properties of complex limits, a sequence of complex numbers converges if and only if their real and imaginary components converge independently. Thus, for every $x \in X$, we have:

$$u(x) = \lim_{j \rightarrow \infty} u_j(x) \quad \text{and} \quad v(x) = \lim_{j \rightarrow \infty} v_j(x)$$

Now we apply our results from the above Proposition:

- (i) Since u is the pointwise limit of a sequence of real-valued measurable functions $\{u_j\}$, $u = \operatorname{Re} f$ is $\mathcal{M}$ -measurable.
- (ii) Since v is the pointwise limit of a sequence of real-valued measurable functions $\{v_j\}$, $v = \operatorname{Im} f$ is $\mathcal{M}$ -measurable.

Finally, by applying Corollary 8 in the forward direction, since both $\operatorname{Re} f$ and $\operatorname{Im} f$ are proven to be $\mathcal{M}$ -measurable, the joint complex limit function f is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$ -measurable. $\square$

Definition 30 (Positive and Negative Parts of a Function). Let $(X, \mathcal{M})$ be a measurable space and let $f : X \rightarrow \overline{\mathbb{R}}$ be an extended real-valued function. The **positive part** of f , denoted by f^+ , and the **negative part** of f , denoted by f^- , are non-negative functions on X defined pointwise by:

$$f^+(x) = \max(f(x), 0)$$

$$f^-(x) = \max(-f(x), 0)$$

Remark. This decomposition is one of the most powerful tools in integration theory. It allows us to split any real-valued function—which might oscillate between positive and negative values—into two purely non-negative pieces.

Here are the key insights your students should keep in mind:

- **Behavior at a point x :**

- If $f(x) \geq 0$, then $f^+(x) = f(x)$ and $f^-(x) = 0$.
- If $f(x) < 0$, then $f^+(x) = 0$ and $f^-(x) = -f(x)$ (which is a positive number).

- **Reconstructing the Function:** The original function and its absolute value can be easily recovered by combining these two parts:

$$f(x) = f^+(x) - f^-(x)$$

$$|f(x)| = f^+(x) + f^-(x)$$

- **Measurability Inheritance:** Since f is measurable, the constant function 0 is measurable, and the scalar multiplication $-f$ is measurable. By Corollary 9, the maximum of two measurable functions is measurable. Therefore, if f is a measurable function, its positive part f^+ and negative part f^- are automatically measurable functions.

Example 9. Let $(X, \mathcal{M})$ be a measurable space. If $f : X \rightarrow \mathbb{C}$ is an $\mathcal{M}$ -measurable function, then its absolute value function $|f| : X \rightarrow \mathbb{R}$, defined pointwise by $|f|(x) = |f(x)|$, is also $\mathcal{M}$ -measurable.

Proof. To make this clear for students, we express the function $|f|$ as a composition of two separate mathematical mappings.

Consider the absolute value function on the complex plane, $g : \mathbb{C} \rightarrow \mathbb{R}$, defined by:

$$g(z) = |z|$$

From classical topology, the map g is continuous everywhere on $\mathbb{C}$ because it satisfies the standard triangle inequality $||z_1| - |z_2|| \leq |z_1 - z_2|$.

By Corollary 2.2, since g is a continuous function between topological spaces, it is automatically a Borel measurable mapping. This means g is $(\mathcal{B}_{\mathbb{C}}, \mathcal{B}_{\mathbb{R}})$ -measurable.

Now, observe that our function of interest, $|f|$, can be written perfectly as the composition of g and f :

$$|f|(x) = |f(x)| = g(f(x)) = (g \circ f)(x)$$

We are given that $f : X \rightarrow \mathbb{C}$ is $\mathcal{M}$ -measurable (which means $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$ -measurable), and we have shown that $g : \mathbb{C} \rightarrow \mathbb{R}$ is $(\mathcal{B}_{\mathbb{C}}, \mathcal{B}_{\mathbb{R}})$ -measurable. Since the composition of two measurable mappings is always measurable, it follows that $|f| = g \circ f$ is an $\mathcal{M}$ -measurable function. $\square$

Example 10. Let $(X, \mathcal{M})$ be a measurable space. If $f : X \rightarrow \mathbb{C}$ is an $\mathcal{M}$ -measurable function, then its complex signum component function $\operatorname{sgn} f : X \rightarrow \mathbb{C}$, defined pointwise by:

$$(\operatorname{sgn} f)(x) = \operatorname{sgn}(f(x)) \quad \text{where} \quad \operatorname{sgn} z = \begin{cases} \frac{z}{|z|} & \text{if } z \neq 0, \\ 0 & \text{if } z = 0, \end{cases}$$

is also $\mathcal{M}$ -measurable.

Proof. Unlike the absolute value function, the standalone complex signum function $\text{sgn} : \mathbb{C} \rightarrow \mathbb{C}$ is *not* continuous on the entire complex plane because it has a sharp jump discontinuity at the origin $z = 0$. Therefore, we cannot invoke Corollary 2.2 directly on $\mathbb{C}$. Instead, we must prove that sgn is Borel measurable by looking at the preimages of open sets.

Let $U \subset \mathbb{C}$ be an arbitrary open target set in the complex plane. We analyze the behavior of the preimage $\text{sgn}^{-1}(U)$ under two distinct conditions based on whether the origin is inside our target window:

1. **Case 1:** $0 \notin U$

If the target open set U does not contain 0, then any point z that maps into U must satisfy $z \neq 0$. On the punctured plane $\mathbb{C} \setminus \{0\}$, the mapping $z \mapsto \frac{z}{|z|}$ is a continuous algebraic combination of continuous functions. Because the restriction of sgn to $\mathbb{C} \setminus \{0\}$ is continuous, the preimage of the open set U must be completely open in $\mathbb{C} \setminus \{0\}$. Since $\mathbb{C} \setminus \{0\}$ is itself an open subset of $\mathbb{C}$, $\text{sgn}^{-1}(U)$ is open in $\mathbb{C}$. Since every open set is a Borel set, $\text{sgn}^{-1}(U) \in \mathcal{B}_{\mathbb{C}}$.

2. **Case 2:** $0 \in U$

If the target open set U contains 0, then by definition, the origin $z = 0$ is pulled back into our preimage set because $\text{sgn}(0) = 0 \in U$. We can split the preimage into the origin piece and everything else:

$$\text{sgn}^{-1}(U) = \text{sgn}^{-1}(U \setminus \{0\}) \cup \{0\}$$

Let $V = \text{sgn}^{-1}(U \setminus \{0\})$. Since $U \setminus \{0\}$ does not contain 0, it falls under the exact logic of **Case 1**, meaning V is an open set in $\mathbb{C}$. Therefore, the total preimage can be written as:

$$\text{sgn}^{-1}(U) = V \cup \{0\}$$

where V is open. Since open sets are Borel sets ($V \in \mathcal{B}_{\mathbb{C}}$) and singleton points are closed and thus Borel sets ($\{0\} \in \mathcal{B}_{\mathbb{C}}$), their union must belong to the Borel σ -algebra $\mathcal{B}_{\mathbb{C}}$.

In both cases, the preimage of an arbitrary open set U under sgn belongs to $\mathcal{B}_{\mathbb{C}}$. By Proposition 2.1, because the open sets generate the Borel σ -algebra, the mapping $\text{sgn} : \mathbb{C} \rightarrow \mathbb{C}$ is $(\mathcal{B}_{\mathbb{C}}, \mathcal{B}_{\mathbb{C}})$ -measurable.

Finally, since $\text{sgn} \circ f$ is the composition of the Borel measurable function sgn with the $\mathcal{M}$ -measurable function f :

$$\text{sgn} \circ f = \text{sgn} \circ f$$

the composition of these two measurable mappings ensures that $\text{sgn} \circ f$ is $\mathcal{M}$ -measurable. $\square$

Definition 31 (Characteristic Function / Indicator Function). Let $(X, \mathcal{M})$ be a measurable space and let $E \subset X$. The **characteristic function** (also known as the **indicator function**) of the set E , denoted by χ_E or 1_E , is a function $\chi_E : X \rightarrow \{0, 1\}$ defined pointwise by:

$$\chi_E(x) = \begin{cases} 1 & \text{if } x \in E, \\ 0 & \text{if } x \notin E. \end{cases}$$

An essential fact for students to understand is that χ_E is an $\mathcal{M}$ -measurable function if and only if the underlying set E is a measurable set (i.e., $E \in \mathcal{M}$).

Definition 32 (Simple Function). A function $f : X \rightarrow \mathbb{C}$ is called a **simple function** if its range is a finite subset of the complex numbers $\mathbb{C}$. Structurally, any such measurable function can be expressed as a finite linear combination, with complex coefficients, of the characteristic functions of sets belonging to our σ -algebra $\mathcal{M}$:

$$f = \sum_{j=1}^n z_j \chi_{E_j}$$

where $z_j \in \mathbb{C}$ and $E_j \in \mathcal{M}$ for all $j \in \{1, \dots, n\}$.

Definition 33 (Standard Representation of a Simple Function). Let $f : X \rightarrow \mathbb{C}$ be a simple function whose distinct non-zero and zero values in its finite range are given by the set $\{z_1, z_2, \dots, z_n\}$. The **standard representation** of f is the unique configuration:

$$f = \sum_{j=1}^n z_j \chi_{E_j}$$

where the sets E_j are explicitly defined as the preimages of each distinct value in the range:

$$E_j = f^{-1}(\{z_j\})$$

In this standard representation, the coefficients z_j are strictly distinct, and the corresponding measurable sets E_j are pairwise disjoint and collectively exhaust the domain space, meaning $\bigcup_{j=1}^n E_j = X$.

Remark (Crucial Pedagogical Notes for Integration Theory). For students transitioning into the theory of Lebesgue integration, the following structural constraints placed on simple functions are highly important:

1. **Measurability of Underlying Sets and Functions:** In our pedagogical context, whenever we define a simple function as a linear combination $\sum z_j \chi_{E_j}$, we implicitly require every single component set E_j to be a measurable set ($E_j \in \mathcal{M}$). This crucial requirement guarantees that the simple function f is strictly measurable. This is foundational because when we later define the Lebesgue integral of a simple function, the definition relies on computing the measures of these individual sets:

$$\int_X f \, d\mu = \sum_{j=1}^n z_j \mu(E_j)$$

If the sets E_j were not elements of $\mathcal{M}$, the measure $\mu(E_j)$ would be completely undefined, breaking our integration framework.

2. **Exclusion of Infinite Values:** By definition, simple functions are strictly maps into the complex numbers $\mathbb{C}$ or real numbers $\mathbb{R}$, and **are not allowed to assume the values $\pm\infty$** . Keeping the range of simple functions entirely finite prevents indeterminate mathematical forms (such as $\infty \times \infty$) from appearing when computing simple linear combination steps during integration.
3. **Domain Exhaustion and Zero Coefficients:** For a simple function to satisfy the basic set-theoretic definition of a function on X , it must assign a clear value to every single element $x \in X$. Consequently, in its standard representation, the union of the disjoint sets E_j must equal the full domain space X ($\bigcup_{j=1}^n E_j = X$).

To achieve this total coverage, **one of the coefficients z_j may well be 0**. Students must not discard the term $0 \cdot \chi_{E_j}$ even though it evaluates to zero; it must be kept as an active structural part of the standard representation because that partition set E_j represents exactly where the function vanishes, and it plays a vital role when f interacts with or is restricted by other functions.

Example 11. Let $(X, \mathcal{M})$ be a measurable space. If $f, g : X \rightarrow \mathbb{C}$ are simple functions, then their sum $f + g$ and their product fg are also simple functions.

Proof. To make this proof completely transparent for students, we must show two things for each combination: first, how the functions look explicitly when combined on a shared partition of the domain, and second, that their ranges remain strictly finite.

Step 1: Constructing a Joint Partition of the Domain Space

By definition, f and g are simple functions, meaning they have finite ranges and can be written in their standard representations:

$$f = \sum_{j=1}^n z_j \chi_{E_j} \quad \text{and} \quad g = \sum_{k=1}^m w_k \chi_{F_k}$$

where $\{z_1, \dots, z_n\} \subset \mathbb{C}$ is the finite range of f , and $\{w_1, \dots, w_m\} \subset \mathbb{C}$ is the finite range of g . The sets $E_j = f^{-1}(\{z_j\})$ partition X , and the sets $F_k = g^{-1}(\{w_k\})$ also partition X . All sets E_j and F_k belong to the σ -algebra $\mathcal{M}$.

To analyze $f + g$ and fg together, we construct a new, finer joint partition of X by intersecting these two families of sets. For every pair of indices (j, k) , define:

$$A_{j,k} = E_j \cap F_k$$

Since $\mathcal{M}$ is a σ -algebra, it is closed under finite intersections, meaning each $A_{j,k} \in \mathcal{M}$. Furthermore, because $\{E_j\}$ and $\{F_k\}$ are individual partitions of X , the collection of all non-empty intersections $\{A_{j,k} : 1 \leq j \leq n, 1 \leq k \leq m\}$ forms a pairwise disjoint partition that perfectly exhausts the full domain space X :

$$\bigcup_{j=1}^n \bigcup_{k=1}^m A_{j,k} = \bigcup_{j=1}^n \bigcup_{k=1}^m (E_j \cap F_k) = \left(\bigcup_{j=1}^n E_j \right) \cap \left(\bigcup_{k=1}^m F_k \right) = X \cap X = X$$

On any specific sub-set $A_{j,k}$, if an element $x \in A_{j,k}$, then $x \in E_j$ (forcing $f(x) = z_j$) and $x \in F_k$ (forcing $g(x) = w_k$).

Step 2: Analysis of the Sum Function ($f + g$)

Let us observe how the sum function looks. By evaluating $f(x) + g(x)$ across our joint partition, we can express the entire sum function as a linear combination over the sets $A_{j,k}$:

$$(f + g)(x) = \sum_{j=1}^n \sum_{k=1}^m (z_j + w_k) \chi_{A_{j,k}}(x)$$

Checking the criteria for a simple function:

1. **Measurability of the component sets:** Every underlying subset $A_{j,k} = E_j \cap F_k$ belongs to $\mathcal{M}$. Thus, $\chi_{A_{j,k}}$ is a measurable indicator function for each term.
2. **Finiteness of the range:** The coefficients in this new expression are of the form $(z_j + w_k)$. Since there are exactly n values for z_j and m values for w_k , the total number of distinct possible values in the range of $f + g$ is at most $n \times m$, which is strictly finite.

Because $f + g$ is a finite linear combination of characteristic functions of measurable sets, it is an $\mathcal{M}$ -measurable function with a finite range in $\mathbb{C}$, matching the definition of a simple function.

Step 3: Analysis of the Product Function (fg)

Similarly, let us observe how the product function looks when evaluated over our joint partition sets. For any point $x \in A_{j,k}$, the product multiplies their respective constant values:

$$(fg)(x) = f(x) \cdot g(x) = z_j \cdot w_k$$

This allows us to write the total product function explicitly as:

$$(fg)(x) = \sum_{j=1}^n \sum_{k=1}^m (z_j w_k) \chi_{A_{j,k}}(x)$$

Checking the criteria for a simple function:

1. **Measurability of the component sets:** Just as before, each set $A_{j,k}$ is a measurable set in $\mathcal{M}$, ensuring that every component function $\chi_{A_{j,k}}$ is measurable.
2. **Finiteness of the range:** The elements making up the range of fg are formed by the products $(z_j w_k)$. The maximum number of distinct values this product range can hold is bounded above by $n \times m$, which is strictly finite.

Since fg can be written as a finite linear combination of indicator functions belonging to $\mathcal{M}$ and does not assume infinite values, it is a simple function.

Note on Standard Representation

Note to students: The joint expressions derived above,

$$\sum_{j,k} (z_j + w_k) \chi_{A_{j,k}} \quad \text{and} \quad \sum_{j,k} (z_j w_k) \chi_{A_{j,k}}$$

are structurally valid linear combinations, but they might not be the official **standard representation** of $f + g$ or fg . If different index pairs yield identical values (for instance, if $z_1 + w_2 = z_3 + w_1$), those matching coefficients must be grouped together, and their corresponding disjoint sets $A_{j,k}$ must be combined using unions to yield the unique standard representation where all coefficients are strictly distinct. Regardless of this final consolidation step, the functions are proved to be simple. $\square$

Theorem 9. *a. If $f : X \rightarrow [0, \infty]$ is measurable, there is a sequence $\{\phi_n\}$ of simple functions such that $0 \leq \phi_1 \leq \phi_2 \leq \dots \leq f$, $\phi_n \rightarrow f$ pointwise, and $\phi_n \rightarrow f$ uniformly on any set on which f is bounded.*

b. If $f : X \rightarrow \mathbb{C}$ is measurable, there is a sequence $\{\phi_n\}$ of simple functions such that $0 \leq |\phi_1| \leq |\phi_2| \leq \dots \leq |f|$, $\phi_n \rightarrow f$ pointwise, and $\phi_n \rightarrow f$ uniformly on any set on which f is bounded.

Proof. To make this challenging proof completely transparent for your students, we will break the entire explanation into structured, digestible phases.

Part (a): Non-negative Extended Functions ($f : X \rightarrow [0, \infty]$)

The core strategy here is to approximate the codomain $[0, \infty]$ by chopping it into finite horizontal strips, seeing where the function lands, and building a simple step-function from the preimages. As n increases, our grid gets taller and finer.

Phase 1: Step-by-Step Evolution of the Construction

Let us build the approximating functions ϕ_n explicitly for the first few stages to see the pattern.

Stage 1: Building ϕ_0

We look at the codomain and establish a ceiling at $2^0 = 1$.

- **The Cutoff:** We divide the codomain into two primary zones: values below $2^0 = 1$, and values above or equal to 1.
- **The Grid Resolution:** For the region $[0, 1)$, we partition it into intervals of width $\frac{1}{2^0} = 1$. This gives us exactly one interval: $[0, 1)$.
- **The Assignment Rule:**
 - If $f(x) \in [0, 1)$, we assign $\phi_0(x) = 0$ (the lower bound of the strip).
 - If $f(x) \in [1, \infty]$, we assign $\phi_0(x) = 1 = 2^0$ (our current ceiling).

Stage 2: Building ϕ_1

Now we double our height ceiling to $2^1 = 2$, and sharpen our grid resolution to step sizes of $\frac{1}{2^1} = \frac{1}{2}$.

- **The Cutoff:** The infinite tail now starts at $2^1 = 2$. If $f(x) \geq 2$, we set $\phi_1(x) = 2$.
- **The Grid Resolution:** The bounded region $[0, 2)$ is chopped into steps of width $\frac{1}{2}$. This yields $2 \times 2^1 = 4$ strips:

$$\left[0, \frac{1}{2}\right), \quad \left[\frac{1}{2}, 1\right), \quad \left[1, \frac{3}{2}\right), \quad \left[\frac{3}{2}, 2\right)$$

- **The Assignment Rule:** For any point x , if $f(x)$ lands in one of these bounded intervals, $\phi_1(x)$ takes the value of the lower edge of that interval (e.g., if $f(x) \in [\frac{1}{2}, 1)$, then $\phi_1(x) = \frac{1}{2}$).

Stage 3: Building ϕ_2

We raise the ceiling to $2^2 = 4$, and make the grid resolution twice as fine: step sizes of $\frac{1}{2^2} = \frac{1}{4}$.

- **The Cutoff:** If $f(x) \geq 4$, we set $\phi_2(x) = 4$.
- **The Grid Resolution:** The bounded range $[0, 4)$ is sliced into sub-intervals of width $\frac{1}{4}$. This creates $4 \times 2^2 = 16$ distinct horizontal strips.
- **The Assignment Rule:** $\phi_2(x)$ tracks the lower endpoint of whichever of the 16 strips $f(x)$ occupies.

General Stage: Building ϕ_n

Following this exact progression, for any arbitrary integer n , our ceiling becomes 2^n and our resolution width shrinks to $\frac{1}{2^n}$. The total number of bounded horizontal slices is $2^n \times 2^n = 2^{2n}$.

Let us formally define the preimages of these slices in the domain space X :

- For the bounded strips, where the index k runs from 0 up to $2^{2n} - 1$:

$$E_n^k = f^{-1} \left(\left[\frac{k}{2^n}, \frac{k+1}{2^n} \right] \right)$$

- For the unbounded tail region beyond our ceiling:

$$F_n = f^{-1}((2^n, \infty])$$

Since f is given to be a measurable function, the preimages of these Borel intervals are guaranteed to be measurable sets in X ($E_n^k, F_n \in \mathcal{M}$).

By multiplying the characteristic indicator functions of these sets by their corresponding lower-edge values, we arrive at the general formula for ϕ_n :

$$\phi_n = \sum_{k=0}^{2^{2n}-1} \frac{k}{2^n} \chi_{E_n^k} + 2^n \chi_{F_n}$$

Because this is a finite linear combination of characteristic functions of measurable sets, ϕ_n is a mathematically valid simple function for every n .

Phase 2: Verifying the Monotone Growth ($\phi_n \leq \phi_{n+1}$)

To show that our sequence grows monotonically towards f , let us look at what happens when we move from stage n to stage $n+1$.

When moving to stage $n+1$, the grid spacing cuts exactly in half, from $\frac{1}{2^n}$ to $\frac{1}{2^{n+1}}$. This means each single horizontal strip from stage n is split perfectly into two sub-strips at stage $n+1$:

$$\text{Strip } k \text{ of stage } n : \left[\frac{k}{2^n}, \frac{k+1}{2^n} \right) \implies \left[\frac{2k}{2^{n+1}}, \frac{2k+1}{2^{n+1}} \right) \cup \left[\frac{2k+1}{2^{n+1}}, \frac{2k+2}{2^{n+1}} \right)$$

Now let us check the value of our function at a point x :

- If $f(x)$ lands in the bottom half of the old strip, its assigned lower-bound value remains exactly the same: $\phi_{n+1}(x) = \frac{2k}{2^{n+1}} = \frac{k}{2^n} = \phi_n(x)$.
- If $f(x)$ lands in the top half of the old strip, its new lower-bound value steps upward: $\phi_{n+1}(x) = \frac{2k+1}{2^{n+1}} > \frac{k}{2^n} = \phi_n(x)$.
- If $f(x)$ was originally trapped by the ceiling 2^n (meaning $\phi_n(x) = 2^n$), the new ceiling at stage $n+1$ has moved up to 2^{n+1} , allowing $\phi_{n+1}(x)$ to measure higher values up to 2^{n+1} .

In all possible scenarios, the value never drops. Thus, $0 \leq \phi_1 \leq \phi_2 \leq \dots \leq f$ holds for all $x \in X$.

Phase 3: Setting Up the Error Bound ($0 \leq f - \phi_n \leq 2^{-n}$)

By design, if the value of our function $f(x)$ is bounded below our ceiling 2^n , it must fall into exactly one of the finite strips $[\frac{k}{2^n}, \frac{k+1}{2^n})$.

On that strip, our simple function rounds the value down to the nearest lower edge: $\phi_n(x) = \frac{k}{2^n}$. Since the total width of this strip is precisely $\frac{1}{2^n}$, the absolute error between the true value $f(x)$ and our rounded value $\phi_n(x)$ cannot exceed the width of that strip. Therefore:

$$\text{If } f(x) \leq 2^n, \quad \text{then } 0 \leq f(x) - \phi_n(x) \leq \frac{1}{2^n} = 2^{-n}$$

—

Phase 4: Proving Pointwise Convergence

Let $x \in X$ be any fixed arbitrary point in our domain. We consider two possibilities for the true value of $f(x)$:

- **Case 1: $f(x)$ is finite ($f(x) < \infty$)**

Since $f(x)$ is a finite number, by the Archimedean property of real numbers, we can choose a large integer N such that $2^N \geq f(x)$. For all indices $n \geq N$, the ceiling 2^n has surpassed the value of $f(x)$. We can then confidently apply our error bound equation from Phase 3:

$$0 \leq f(x) - \phi_n(x) \leq \frac{1}{2^n} \quad \text{for all } n \geq N$$

Taking the limit as $n \rightarrow \infty$, the squeeze theorem forces the error to zero:

$$\lim_{n \rightarrow \infty} (f(x) - \phi_n(x)) = 0 \implies \lim_{n \rightarrow \infty} \phi_n(x) = f(x)$$

- **Case 2: $f(x)$ is infinite ($f(x) = \infty$)**

If $f(x) = \infty$, then for every single index n , the value of $f(x)$ always sits past our ceiling 2^n . According to our construction rule, if a point lies in the tail F_n , the function outputs the ceiling value:

$$\phi_n(x) = 2^n \quad \text{for all } n$$

Taking the limit of this sequence gives:

$$\lim_{n \rightarrow \infty} \phi_n(x) = \lim_{n \rightarrow \infty} 2^n = \infty = f(x)$$

Thus, $\phi_n \rightarrow f$ pointwise across the entire domain space X .

—

Phase 5: Proving Uniform Convergence on Bounded Sets

Let $B \subset X$ be any subset of our domain where the function f is bounded. This means there exists some fixed finite constant $M > 0$ such that:

$$f(x) \leq M \quad \text{for all } x \in B$$

Let us choose a fixed integer N large enough such that $2^N \geq M$. This means that for any index $n \geq N$, the entire set B fits completely below our moving ceiling ($f(x) \leq M \leq 2^N \leq 2^n$).

Because every single point $x \in B$ lies below the ceiling for any $n \geq N$, the uniform error bound derived in Phase 3 applies globally across the entire set B at the same time:

$$\sup_{x \in B} |f(x) - \phi_n(x)| \leq \frac{1}{2^n} \quad \text{for all } n \geq N$$

Since $\frac{1}{2^n} \rightarrow 0$ as $n \rightarrow \infty$, the supremum of the error over the entire set B vanishes. This proves that $\phi_n \rightarrow f$ uniformly on B .

—

Part (b): Complex-Valued Functions ($f : X \rightarrow \mathbb{C}$)

Now we extend our proof to a complex-valued measurable function f . We do this by breaking the complex function down into its real component building blocks, applying part (a), and recombining them.

Step 1: Splitting into Non-negative Real Functions

Let us decompose f into its real and imaginary parts:

$$f(x) = g(x) + ih(x) \quad \text{where } g = \operatorname{Re} f \text{ and } h = \operatorname{Im} f$$

By Corollary 2.5, since f is complex-measurable, both g and h are real-valued measurable functions on X .

Next, we split these real functions into their corresponding positive and negative parts:

$$g(x) = g^+(x) - g^-(x) \quad \text{and} \quad h(x) = h^+(x) - h^-(x)$$

By our earlier work on functional decompositions, all four functions— g^+ , g^- , h^+ , and h^- —are non-negative, finite, measurable functions on X .

Step 2: Constructing Approximating Sequences for Each Component

Since g^+, g^-, h^+, h^- are non-negative and measurable, we can apply the results from **Part (a)** to each component individually. This yields four independent sequences of non-negative simple functions that climb monotonically to their targets:

$$\psi_n^+ \nearrow g^+, \quad \psi_n^- \nearrow g^-, \quad \zeta_n^+ \nearrow h^+, \quad \zeta_n^- \nearrow h^-$$

Now, we combine these four separate simple sequences back into a single complex-valued sequence $\{\phi_n\}$ defined by:

$$\phi_n = (\psi_n^+ - \psi_n^-) + i(\zeta_n^+ - \zeta_n^-)$$

Because the sum, difference, and scalar multiplication of simple functions yield simple functions, ϕ_n is a complex simple function for every n .

Step 3: Verifying Pointwise and Uniform Convergence

Since limits commute with basic linear combinations:

$$\begin{aligned} \lim_{n \rightarrow \infty} \phi_n &= \left(\lim_{n \rightarrow \infty} \psi_n^+ - \lim_{n \rightarrow \infty} \psi_n^- \right) + i \left(\lim_{n \rightarrow \infty} \zeta_n^+ - \lim_{n \rightarrow \infty} \zeta_n^- \right) \\ &= (g^+ - g^-) + i(h^+ - h^-) = g + ih = f \end{aligned}$$

This proves $\phi_n \rightarrow f$ pointwise. Furthermore, if f is bounded on a set B , then its components g and h must be bounded on B , forcing their positive and negative parts to be bounded. Since each component sequence converges uniformly on B by Part (a), their linear combination ϕ_n converges uniformly to f on B .

Step 4: Proving Monotone Growth of the Absolute Values ($|\phi_n| \leq |\phi_{n+1}|$)

To complete the proof, we must show that the magnitudes grow monotonically: $|\phi_n| \leq |\phi_{n+1}| \leq |f|$.

Recall that for our constructed sequences, at any point x , the functions $\psi_n^+(x)$ and $\psi_n^-(x)$ cannot be non-zero at the same time because $g^+(x)$ and $g^-(x)$ are disjointly supported. Therefore, the absolute value of the real part simplifies cleanly:

$$|\operatorname{Re} \phi_n| = |\psi_n^+ - \psi_n^-| = \psi_n^+ + \psi_n^-$$

Because $\psi_n^+ \leq \psi_{n+1}^+$ and $\psi_n^- \leq \psi_{n+1}^-$, it follows that:

$$|\operatorname{Re} \phi_n| \leq |\operatorname{Re} \phi_{n+1}| \leq \operatorname{Re} f^+ + \operatorname{Re} f^- = |g|$$

By identical logic applied to the imaginary components:

$$|\operatorname{Im} \phi_n| \leq |\operatorname{Im} \phi_{n+1}| \leq |h|$$

Using the definition of the complex modulus ($|z| = \sqrt{(\operatorname{Re} z)^2 + (\operatorname{Im} z)^2}$), since the absolute values of both components grow monotonically at every step, the total magnitude must grow monotonically:

$$|\phi_n| = \sqrt{|\operatorname{Re} \phi_n|^2 + |\operatorname{Im} \phi_n|^2} \leq \sqrt{|\operatorname{Re} \phi_{n+1}|^2 + |\operatorname{Im} \phi_{n+1}|^2} = |\phi_{n+1}|$$

And since components are bounded by their targets:

$$|\phi_{n+1}| \leq \sqrt{|g|^2 + |h|^2} = |f|$$

This establishes $0 \leq |\phi_1| \leq |\phi_2| \leq \dots \leq |f|$, completes the proof. $\square$

Proposition (Proposition 2.11). *The following implications are valid iff the measure μ is complete:*

- a. *If f is measurable and $f = g$ μ -a.e., then g is measurable.*
- b. *If f_n is measurable for $n \in \mathbb{N}$ and $f_n \rightarrow f$ μ -a.e., then f is measurable.*

Proof. To make this proof fully transparent for your students, we will split our argument into two directional halves: the forward direction ($\implies$), where we assume completeness to prove statements (a) and (b), and the backward direction ($\impliedby$), where we show that if either statement holds, the measure space must be complete.

Direction 1: Assume μ is Complete ($\implies$)

Let $(X, \mathcal{M}, \mu)$ be a complete measure space. Recall that a measure space is **complete** if every subset of a set of measure zero is itself measurable (i.e., if $N \in \mathcal{M}$ with $\mu(N) = 0$, then any $E \subset N$ satisfies $E \in \mathcal{M}$, which forces $\mu(E) = 0$).

Proof of Part (a)

We are given that f is an $\mathcal{M}$ -measurable function, and $f = g$ μ -almost everywhere (μ -a.e.).

Step 1: Isolate the exceptional set: By definition of μ -a.e., the set of points where f and g differ is contained within a measurable set of measure zero. Let:

$$N = \{x \in X : f(x) \neq g(x)\}$$

Since $f = g$ μ -a.e., there exists a set $M \in \mathcal{M}$ such that $N \subset M$ and $\mu(M) = 0$.

Step 2: Analyze preimages under g : To prove that g is measurable, we must show that for any Borel set $B \in \mathcal{B}_{\mathbb{R}}$ (or $\mathcal{B}_{\overline{\mathbb{R}}}$), its preimage under g is a member of $\mathcal{M}$. Let us partition the preimage $g^{-1}(B)$ into two distinct parts: points outside our exceptional set M , and points inside M .

$$g^{-1}(B) = \left(g^{-1}(B) \cap M^c\right) \cup \left(g^{-1}(B) \cap M\right)$$

Step 3: Evaluate each component:

- **Outside the exceptional set (M^c):** On the set M^c , the functions f and g are identical. Therefore, $g^{-1}(B) \cap M^c = f^{-1}(B) \cap M^c$. Because f is measurable, $f^{-1}(B) \in \mathcal{M}$. Since $M \in \mathcal{M}$, its complement $M^c \in \mathcal{M}$. The intersection of two measurable sets is measurable, so $(f^{-1}(B) \cap M^c) \in \mathcal{M}$.
- **Inside the exceptional set (M):** Let $E = g^{-1}(B) \cap M$. By construction, E is a subset of M ($E \subset M$). Since our measure space is **complete** and M is a measurable set of measure zero ($\mu(M) = 0$), the subset E is automatically guaranteed to be a measurable set ($E \in \mathcal{M}$).

Step 4: Recombine the components: We can now express $g^{-1}(B)$ as the union of two verified elements of $\mathcal{M}$:

$$g^{-1}(B) = \underbrace{\left(f^{-1}(B) \cap M^c\right)}_{\in \mathcal{M}} \cup \underbrace{\left(g^{-1}(B) \cap M\right)}_{\in \mathcal{M}}$$

Since $\mathcal{M}$ is a σ -algebra, it is closed under countable unions. Therefore, $g^{-1}(B) \in \mathcal{M}$ for any Borel set B , proving that g is an $\mathcal{M}$ -measurable function.

Proof of Part (b)

We are given a sequence $\{f_n\}$ of measurable functions such that $f_n \rightarrow f$ μ -a.e..

Step 1: Isolate the non-convergence set: Let $N \subset X$ be the set of points where the sequence $f_n(x)$ fails to converge to $f(x)$. Since $f_n \rightarrow f$ μ -a.e., there exists a measurable set $M \in \mathcal{M}$ such that $N \subset M$ and $\mu(M) = 0$.

Step 2: Construct a modified sequence: Let us define a new sequence of functions $\{g_n\}$ that forces convergence everywhere by zeroing out the sequence on our exceptional set M :

$$g_n(x) = f_n(x)\chi_{M^c}(x) = \begin{cases} f_n(x) & \text{if } x \notin M \\ 0 & \text{if } x \in M \end{cases}$$

Because each f_n is measurable and M^c is a measurable set, the characteristic function χ_{M^c} is measurable. The product of two measurable functions is measurable, so each g_n is strictly an $\mathcal{M}$ -measurable function.

Step 3: Analyze the pointwise limit of g_n : Because we explicitly zeroed out the sequence on M , the sequence $g_n(x)$ converges pointwise at *every single point* in X to a limit function $g(x)$, defined by:

$$g(x) = \lim_{n \rightarrow \infty} g_n(x) = \begin{cases} f(x) & \text{if } x \notin M \\ 0 & \text{if } x \in M \end{cases}$$

By Proposition 2.7 (or Corollary 2.9), the pointwise limit of a sequence of measurable functions is always measurable. Therefore, g is a measurable function.

Step 4: Link g back to f : By comparing the definitions of $f(x)$ and $g(x)$, we see that they are identical everywhere except possibly on our exceptional set M . Because M has measure zero, we have:

$$f = g \quad \mu\text{-a.e.}$$

We have already established that g is a measurable function. By applying our result from **Part (a)**, since f equals a measurable function g μ -almost everywhere, f is guaranteed to be measurable.

—

Direction 2: Prove Completeness is Necessary ($\Leftarrow$)

To establish the "iff" condition, we must show that if $(X, \mathcal{M}, \mu)$ is *not* complete, then implications (a) and (b) can fail. We do this by demonstrating that if statement (a) holds, the space must be complete.

Suppose statement (a) holds valid on our measure space. Let $M \in \mathcal{M}$ be an arbitrary set with $\mu(M) = 0$, and let $E \subset M$ be an arbitrary subset of M . To prove completeness, we must show that $E \in \mathcal{M}$.

Step 1: Define a tracking function for E : Let $g = \chi_E$ be the characteristic indicator function of our subset E :

$$g(x) = \begin{cases} 1 & \text{if } x \in E \\ 0 & \text{if } x \notin E \end{cases}$$

Step 2: Define a baseline comparison function: Let $f = 0$ be the zero function on X . The constant function 0 is trivially measurable.

Step 3: Show that $f = g$ μ -almost everywhere: Let us find the set of points where $f(x) \neq g(x)$:

$$\{x \in X : f(x) \neq g(x)\} = \{x \in X : 0 \neq \chi_E(x)\} = E$$

By our initial choice, $E \subset M$ and $\mu(M) = 0$. Therefore, the set of points where f and g differ is contained inside a set of measure zero. This satisfies the definition of almost everywhere equivalence:

$$f = g \quad \mu\text{-a.e.}$$

Step 4: **Invoke Statement (a):** By our hypothesis, statement (a) is valid. Since f is measurable and $f = g$ μ -a.e., statement (a) forces the function $g = \chi_E$ to be an $\mathcal{M}$ -measurable function.

Step 5: **Conclude completeness:** Recall the fundamental property of indicator functions: a characteristic function χ_E is measurable if and only if its underlying set E belongs to the σ -algebra $\mathcal{M}$. Because $g = \chi_E$ is proven to be measurable, it follows directly that $E \in \mathcal{M}$.

Since an arbitrary subset E of any null set M is a member of $\mathcal{M}$, the measure space $(X, \mathcal{M}, \mu)$ is complete. (Note: An identical argument can be made using statement (b) by setting a constant sequence $f_n = 0$, which converges μ -a.e. to χ_E). $\square$

Proposition (Proposition 2.12). *Let $(X, \mathcal{M}, \mu)$ be a measure space and let $(X, \overline{\mathcal{M}}, \overline{\mu})$ be its completion. If f is an $\overline{\mathcal{M}}$ -measurable function on X , there is an $\mathcal{M}$ -measurable function g such that $f = g$ $\overline{\mu}$ -almost everywhere.*

Proof. To ensure total clarity for students, we will break this proof into three progressive stages: first establishing the result for characteristic functions, expanding it linearly to simple functions, and finally taking the pointwise limit to cover arbitrary measurable functions.

Throughout this proof, we recall the definition of the completion $\overline{\mathcal{M}}$: a set $E \in \overline{\mathcal{M}}$ if and only if it can be decomposed as $E = A \cup E_0$, where $A \in \mathcal{M}$ and $E_0 \subset N_0$ for some $N_0 \in \mathcal{M}$ with $\mu(N_0) = 0$. By definition, $\overline{\mu}(E) = \mu(A)$.

Part 1: The Base Case for Characteristic Functions

Let $f = \chi_E$ be an $\overline{\mathcal{M}}$ -measurable characteristic function, which means that the underlying set E belongs to the completed σ -algebra $\overline{\mathcal{M}}$.

Step 1: **Decompose the completed set:** By the definition of a completion, since $E \in \overline{\mathcal{M}}$, we can write $E = A \cup E_0$, where $A \in \mathcal{M}$ and $E_0 \subset N_0$ for some symmetric null set $N_0 \in \mathcal{M}$ satisfying $\mu(N_0) = 0$.

Step 2: **Construct the target function:** We define our candidate function $g : X \rightarrow \mathbb{R}$ to be the characteristic function of the clean, original measurable set A :

$$g = \chi_A$$

Since $A \in \mathcal{M}$, the function g is strictly $\mathcal{M}$ -measurable.

Step 3: **Verify almost-everywhere equality:** Let us find where $f(x)$ and $g(x)$ fail to match:

$$\{x \in X : \chi_E(x) \neq \chi_A(x)\} = E \setminus A = E_0 \subset N_0$$

Since $N_0 \in \mathcal{M}$ and $\mu(N_0) = 0$, the definition of the completion ensures that $\overline{\mu}(N_0) = \mu(N_0) = 0$. Since the exceptional set is contained within a set of $\overline{\mu}$ -measure zero, we conclude:

$$f = g \quad \overline{\mu}\text{-almost everywhere}$$

Part 2: Extension to Simple Functions

Let f be an $\overline{\mathcal{M}}$ -measurable simple function.

Step 1: **Represent the function:** We can write f using its standard representation:

$$f = \sum_{j=1}^n z_j \chi_{E_j}$$

where $z_j \in \mathbb{C}$ and each component set $E_j \in \overline{\mathcal{M}}$.

Step 2: **Apply the base case term-by-term:** By applying **Part 1** to each individual completed set E_j , we find corresponding sets $A_j \in \mathcal{M}$ such that $E_j \setminus A_j \subset N_j$ for some $N_j \in \mathcal{M}$ with $\mu(N_j) = 0$.

Step 3: **Construct the simple target function:** We piece together these separate components to build a clean function g :

$$g = \sum_{j=1}^n z_j \chi_{A_j}$$

Since every single $A_j \in \mathcal{M}$, the linear combination g is strictly $\mathcal{M}$ -measurable.

Step 4: **Verify almost-everywhere equality:** The functions f and g can only differ if at least one of their component pairs differ. Therefore, the total exceptional set is bounded by the finite union of the individual exceptional sets:

$$\{x \in X : f(x) \neq g(x)\} \subset \bigcup_{j=1}^n (E_j \setminus A_j) \subset \bigcup_{j=1}^n N_j$$

Let $N_{\text{total}} = \bigcup_{j=1}^n N_j$. Since $\mathcal{M}$ is a σ -algebra, $N_{\text{total}} \in \mathcal{M}$. By subadditivity:

$$\mu(N_{\text{total}}) \leq \sum_{j=1}^n \mu(N_j) = \sum_{j=1}^n 0 = 0$$

Since the subset of disagreement is trapped inside a set of $\bar{\mu}$ -measure zero, $f = g$ $\bar{\mu}$ -a.e. holds for all simple functions.

—

Part 3: The General Case via Pointwise Limits

Let f be an arbitrary general $\overline{\mathcal{M}}$ -measurable function on X .

Step 1: **Approximate with simple functions:** By Theorem 2.10, any measurable function can be realized as the pointwise limit of a sequence of simple functions. Therefore, there exists a sequence $\{\phi_n\}$ of $\overline{\mathcal{M}}$ -measurable simple functions such that:

$$\phi_n(x) \rightarrow f(x) \quad \text{pointwise for all } x \in X$$

Step 2: **Replace the sequence components:** By applying our results from **Part 2** to each term in the sequence, for every index n , there exists an $\mathcal{M}$ -measurable simple function ψ_n that matches ϕ_n everywhere except on an exceptional set $E_n \in \overline{\mathcal{M}}$ with $\bar{\mu}(E_n) = 0$. By definition, each E_n is contained within a measurable null set $M_n \in \mathcal{M}$ such that $\mu(M_n) = 0$.

Step 3: **Construct a unified exceptional set N :** We collect all these separate problem sites into a single master set:

$$N = \bigcup_{n=1}^{\infty} M_n$$

Why does this N work?

- Since each $M_n \in \mathcal{M}$ and a σ -algebra is closed under countable unions, our master set N is guaranteed to belong to $\mathcal{M}$.
- By countable subadditivity of measures, $\mu(N) \leq \sum_{n=1}^{\infty} \mu(M_n) = \sum_{n=1}^{\infty} 0 = 0$. Thus, N is a strictly valid measurable set of measure zero.
- Crucially, because N contains every single M_n , it contains all individual exceptional sets ($N \supset \bigcup_{n=1}^{\infty} E_n$). Therefore, outside of this single set N , **every single function ψ_n matches its counterpart ϕ_n perfectly simultaneously.**

Step 4: **Construct the final function g :** We define our final target function g by taking the pointwise limit of our clean sequence, while explicitly forcing it to zero on our master exceptional set to protect against divergence:

$$g(x) = \lim_{n \rightarrow \infty} \chi_{X \setminus N}(x) \psi_n(x)$$

Step 5: **Verify that g is measurable:** Since $N \in \mathcal{M}$, its complement $X \setminus N \in \mathcal{M}$, meaning the indicator function $\chi_{X \setminus N}$ is $\mathcal{M}$ -measurable. Because each ψ_n is $\mathcal{M}$ -measurable, the products are $\mathcal{M}$ -measurable. Finally, by Corollary 2.9, the pointwise limit of a sequence of measurable functions is always measurable. Therefore, g is an $\mathcal{M}$ -measurable function.

Step 6: **Verify that $f = g$ $\bar{\mu}$ -almost everywhere:** Let us evaluate the behavior of our functions on the complement set N^c : For any point $x \in N^c$:

- The indicator function evaluates to $\chi_{X \setminus N}(x) = 1$.
- Since $x \notin N$, it belongs to none of the exceptional sets E_n , meaning $\psi_n(x) = \phi_n(x)$ for every single n .

Substituting these values into the definition of g :

$$g(x) = \lim_{n \rightarrow \infty} (1 \cdot \psi_n(x)) = \lim_{n \rightarrow \infty} \phi_n(x) = f(x)$$

This demonstrates that $g(x) = f(x)$ for every single point in N^c . Because the complement set N has a $\bar{\mu}$ -measure of zero, we have established that:

$$f = g \quad \bar{\mu}\text{-almost everywhere}$$

This completes the proof for the general case. □

2.2 Integration of Nonnegative Functions

Definition 34 (The Space L^+). We fix a measure space $(X, \mathcal{M}, \mu)$. We define:

$$L^+ = \{f : X \rightarrow [0, \infty] \mid f \text{ is } \mathcal{M}\text{-measurable}\}$$

representing the space of all non-negative, extended real-valued measurable functions on X .

Definition 35 (Integral of a Nonnegative Simple Function). If $\phi \in L^+$ is a simple function with its standard representation given by:

$$\phi = \sum_{j=1}^n a_j \chi_{E_j}$$

where $a_j \in [0, \infty)$ are distinct values and the sets $E_j = \phi^{-1}(\{a_j\})$ are pairwise disjoint elements of $\mathcal{M}$ whose union is X , we define the **integral of ϕ with respect to μ** by:

$$\int \phi \, d\mu = \sum_{j=1}^n a_j \mu(E_j)$$

Explanatory Remarks

Remark (The Necessity of the $0 \cdot \infty = 0$ Convention). In measure theory, we strictly enforce the structural convention that:

$$0 \cdot \infty = 0$$

This algebraic rule is necessary because a simple function can take the value 0 on a set of infinite measure. Without this convention, computing $\int \phi \, d\mu$ using the definition above would leave the term $0 \cdot \mu(E_j) = 0 \cdot \infty$ completely undefined, paralyzing our arithmetic on unbounded domains.

Example: Consider the real line with the Lebesgue measure, $(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, m)$. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as:

$$f(x) = \begin{cases} 1 & \text{if } 0 < x \leq 2 \\ 2 & \text{if } 2 < x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

The standard representation of this simple function partitions the entire space $X = \mathbb{R}$ into three disjoint measurable sets:

- $E_1 = (0, 2]$ with value $a_1 = 1$, where $m(E_1) = 2$.
- $E_2 = (2, 4]$ with value $a_2 = 2$, where $m(E_2) = 2$.
- $E_3 = (-\infty, 0] \cup (4, \infty)$ with value $a_3 = 0$, where $m(E_3) = \infty$.

When we calculate the total integral over $\mathbb{R}$ according to our definition:

$$\begin{aligned} \int f \, dm &= a_1 m(E_1) + a_2 m(E_2) + a_3 m(E_3) \\ &= (1 \cdot 2) + (2 \cdot 2) + \underbrace{(0 \cdot \infty)}_{=0} \\ &= 2 + 4 + 0 = 6 \end{aligned}$$

By assigning $0 \cdot \infty = 0$, the infinite tails of the real line where the function is dead drop away cleanly, matching our geometric intuition that the total area under this step curve is finite.

Remark (Integration over Measurable Subsets). If $A \in \mathcal{M}$ is a measurable subset of X , we define the integration of a simple function ϕ over the sub-domain A by restricting the function using the characteristic indicator χ_A :

$$\int_A \phi \, d\mu = \int \phi \chi_A \, d\mu$$

This modification preserves our operational framework because the product $\phi \chi_A$ remains a strictly valid simple function. Given the standard representation $\phi = \sum a_j \chi_{E_j}$, distributing χ_A reveals:

$$\phi \chi_A = \left(\sum_{j=1}^n a_j \chi_{E_j} \right) \chi_A = \sum_{j=1}^n a_j \chi_{E_j \cap A}$$

Because $\mathcal{M}$ is a σ -algebra, each intersection $(E_j \cap A)$ is a measurable set, meaning $\phi \chi_A$ is a finite linear combination of measurable indicator functions—confirming it is simple. Thus, its subset integral can be computed directly as:

$$\int_A \phi \, d\mu = \sum_{j=1}^n a_j \mu(E_j \cap A)$$

Remark (Notational Variations). When there is no risk of ambiguity regarding the underlying measure space, several standard shorthand notations are used interchangeably in analytical literature:

- $\int \phi$: The most minimalist representation, omitting the measure symbol entirely when the background context implies μ .
- $\int \phi(x) \, d\mu(x)$: Used to explicitly track the dummy variable of integration, which becomes helpful when handling compound functions or evaluating nested multiple integrals.
- $\int \phi(x) \, \mu(dx)$: A notation common in probability theory and functional analysis that treats the measure expression as an operator acting on changing differential increments.

Proposition (Proposition 2.13). Let ϕ and ψ be simple functions in L^+ .

- If $c \geq 0$, $\int c\phi = c \int \phi$.
- $\int (\phi + \psi) = \int \phi + \int \psi$.
- If $\phi \leq \psi$, then $\int \phi \leq \int \psi$.
- The map $A \mapsto \int_A \phi \, d\mu$ is a measure on $\mathcal{M}$.

Proof. Detailed Proof of Part (a)

Let $\phi = \sum_{j=1}^n a_j \chi_{E_j}$ be the standard representation of the simple function ϕ , where $a_j \in [0, \infty)$ are the distinct values taken by ϕ , and $E_j = \phi^{-1}(\{a_j\})$ are pairwise disjoint measurable sets whose union is X .

Step 1: Handle the case $c = 0$. If $c = 0$, then the function $c\phi = 0 \cdot \phi$ is the constant zero function on X . The standard representation of the zero function is simply $0 \cdot \chi_X$. By definition of the simple integral:

$$\int 0 \, d\mu = 0 \cdot \mu(X)$$

Under the core conventions of measure theory, if $\mu(X) = \infty$, the product evaluates via $0 \cdot \infty = 0$. Thus, $\int 0 \cdot \phi = 0$. On the other side, $c \int \phi = 0 \cdot \int \phi = 0$. Hence, equality holds when $c = 0$.

Step 2: Handle the case $c > 0$. If $c > 0$, the function $c\phi$ scales every value output. Since the original constants a_j are distinct, the scaled values ca_j remain distinct. The preimage sets are completely unaffected because the set of points where $c\phi(x) = ca_j$ is exactly the set of points where $\phi(x) = a_j$:

$$(c\phi)^{-1}(\{ca_j\}) = \phi^{-1}(\{a_j\}) = E_j$$

Therefore, the standard representation of the new function $c\phi$ is exactly $\sum_{j=1}^n (ca_j)\chi_{E_j}$.

Step 3: Apply the definition of the integral. Using the definition of the simple function integral directly on this standard representation:

$$\int c\phi \, d\mu = \sum_{j=1}^n (ca_j)\mu(E_j)$$

Since c is a fixed real constant factor in each term of this finite algebraic sum, we can factor it out using standard distributive laws:

$$\int c\phi \, d\mu = c \sum_{j=1}^n a_j \mu(E_j) = c \int \phi \, d\mu$$

—

Detailed Proof of Part (b)

Let $\phi = \sum_{j=1}^n a_j \chi_{E_j}$ and $\psi = \sum_{k=1}^m b_k \chi_{F_k}$ be the standard representations of ϕ and ψ respectively. This establishes that $\{E_j\}_{j=1}^n$ is a partition of X and $\{F_k\}_{k=1}^m$ is another partition of X .

Step 1: Set-Theoretic breakdown via common refinement. Because the sets $\{E_j\}$ cover the entire space X , intersecting any set F_k with the entire collection decomposes it into localized, non-overlapping patches:

$$F_k = F_k \cap X = F_k \cap \left(\bigcup_{j=1}^n E_j \right) = \bigcup_{j=1}^n (E_j \cap F_k)$$

Similarly, each set E_j can be decomposed across the partition components of F_k :

$$E_j = E_j \cap X = E_j \cap \left(\bigcup_{k=1}^m F_k \right) = \bigcup_{k=1}^m (E_j \cap F_k)$$

Because these components are pairwise disjoint, applying the property of finite additivity of the measure μ converts these set unions into clear numerical sums:

$$\mu(E_j) = \sum_{k=1}^m \mu(E_j \cap F_k) \quad \text{and} \quad \mu(F_k) = \sum_{j=1}^n \mu(E_j \cap F_k)$$

Step 2: Express independent integrals using the refined sets. We can now expand the definition of $\int \phi$ and $\int \psi$ by substituting these measure decompositions:

$$\int \phi = \sum_{j=1}^n a_j \mu(E_j) = \sum_{j=1}^n a_j \left(\sum_{k=1}^m \mu(E_j \cap F_k) \right) = \sum_{j,k} a_j \mu(E_j \cap F_k)$$

$$\int \psi = \sum_{k=1}^m b_k \mu(F_k) = \sum_{k=1}^m b_k \left(\sum_{j=1}^n \mu(E_j \cap F_k) \right) = \sum_{j,k} b_k \mu(E_j \cap F_k)$$

Adding these two decoupled double sums together allows us to group terms by their shared intersection measure $\mu(E_j \cap F_k)$:

$$\int \phi + \int \psi = \sum_{j,k} a_j \mu(E_j \cap F_k) + \sum_{j,k} b_k \mu(E_j \cap F_k) = \sum_{j,k} (a_j + b_k) \mu(E_j \cap F_k)$$

Step 3: Match with the integral of the sum function. Now let's look directly at the combined function $(\phi + \psi)$. For any arbitrary point $x \in X$, that point must fall into exactly one specific intersection tile $E_j \cap F_k$. At that point, $\phi(x) = a_j$ and $\psi(x) = b_k$, which means:

$$(\phi + \psi)(x) = a_j + b_k \quad \text{for all } x \in E_j \cap F_k$$

Thus, we can represent the combined function as a linear combination over the refinement grid:

$$\phi + \psi = \sum_{j,k} (a_j + b_k) \chi_{E_j \cap F_k}$$

Although this layout may not be a *standard representation* (since different grid tiles might accidentally share the same total value, e.g., $a_j + b_k = a_{j'} + b_{k'}$), grouping identical coefficients together proves that the calculated integral is invariant under refined representations. Calculating the integral from this form yields:

$$\int (\phi + \psi) = \sum_{j,k} (a_j + b_k) \mu(E_j \cap F_k)$$

This matches our final equation from Step 2 identically, completing the proof of additivity.

Detailed Proof of Part (c)

Let $\phi = \sum_{j=1}^n a_j \chi_{E_j}$ and $\psi = \sum_{k=1}^m b_k \chi_{F_k}$ be the standard representations. We are given that $\phi(x) \leq \psi(x)$ for all $x \in X$.

Step 1: Pointwise constraint on overlapping zones. Consider any refined intersection domain $E_j \cap F_k$. If this set is empty, its measure is zero and it does not affect our calculations. If it is non-empty, there is a point $x \in E_j \cap F_k$. At this point, $\phi(x) = a_j$ and $\psi(x) = b_k$. Because $\phi \leq \psi$ everywhere on X , it follows that:

$$a_j \leq b_k \quad \text{whenever } E_j \cap F_k \neq \emptyset$$

Step 2: Comparison of the expanded sums. Using the refinement mesh formulas derived in Part (b), we express both integrals over the identical base sets:

$$\int \phi = \sum_{j,k} a_j \mu(E_j \cap F_k) \quad \text{and} \quad \int \psi = \sum_{j,k} b_k \mu(E_j \cap F_k)$$

Since the scalar measures $\mu(E_j \cap F_k)$ are strictly non-negative, and $a_j \leq b_k$ for every single non-empty tile, we can perform a direct term-by-term inequality comparison:

$$a_j \mu(E_j \cap F_k) \leq b_k \mu(E_j \cap F_k) \quad \text{for all } j, k$$

Summing up these individual inequalities preserves the overall relationship, confirming:

$$\sum_{j,k} a_j \mu(E_j \cap F_k) \leq \sum_{j,k} b_k \mu(E_j \cap F_k) \implies \int \phi \leq \int \psi$$

Detailed Proof of Part (d)

Let $\phi = \sum_{j=1}^n a_j \chi_{E_j}$ be the standard representation of our simple function on X . We define a set function $\nu : \mathcal{M} \rightarrow [0, \infty]$ by $\nu(A) = \int_A \phi d\mu$. To prove that ν constitutes a valid measure, we must verify the two basic axioms of a measure.

Step 1: Verify that the empty set maps to zero. Let $A = \emptyset$. By definition of subset integration over a simple function:

$$\nu(\emptyset) = \sum_{j=1}^n a_j \mu(E_j \cap \emptyset) = \sum_{j=1}^n a_j \mu(\emptyset)$$

Since the underlying μ is a valid measure, $\mu(\emptyset) = 0$. Since all coefficients a_j are finite non-negative numbers, the product $a_j \cdot 0$ is 0 for every term. Thus, $\nu(\emptyset) = 0$.

Step 2: Verify Countable Additivity. Let $\{A_k\}_{k=1}^\infty$ be a sequence of pairwise disjoint sets in $\mathcal{M}$, and denote their union as $A = \bigcup_{k=1}^\infty A_k$. We apply the definition of subset integration to evaluate $\nu(A)$:

$$\nu(A) = \sum_{j=1}^n a_j \mu(A \cap E_j)$$

We expand the set intersection inside the measure expression using distributive laws:

$$A \cap E_j = \left(\bigcup_{k=1}^\infty A_k \right) \cap E_j = \bigcup_{k=1}^\infty (A_k \cap E_j)$$

Because the sets A_k are mutually disjoint, their individual intersections with E_j are also completely disjoint. Since the background measure μ is countably additive, we can rewrite this measure of a countable union as a convergent infinite series:

$$\mu(A \cap E_j) = \sum_{k=1}^\infty \mu(A_k \cap E_j)$$

Substituting this series expression back into our formula for $\nu(A)$ gives:

$$\nu(A) = \sum_{j=1}^n a_j \left(\sum_{k=1}^\infty \mu(A_k \cap E_j) \right) = \sum_{j=1}^n \sum_{k=1}^\infty a_j \mu(A_k \cap E_j)$$

Step 3: Change the summation order. The expression above consists of a finite sum of infinite series containing purely non-negative terms. By standard analytical rules governing series with values in $[0, \infty]$, we can change the order of summation without altering the final value:

$$\nu(A) = \sum_{k=1}^\infty \left(\sum_{j=1}^n a_j \mu(A_k \cap E_j) \right)$$

We recognize that the inner finite sum is exactly the definition of the subset integral over the set A_k :

$$\sum_{j=1}^n a_j \mu(A_k \cap E_j) = \int_{A_k} \phi d\mu = \nu(A_k)$$

Replacing the inner block with $\nu(A_k)$ leaves us with:

$$\nu(A) = \sum_{k=1}^\infty \nu(A_k)$$

This proves countable additivity, establishing that the assignment $A \mapsto \int_A \phi d\mu$ is a fully qualified measure. $\square$

Definition of the Integral for General Nonnegative Functions

We extend the definition of the integral from simple functions to all nonnegative measurable functions $f \in L^+$. If $f : X \rightarrow [0, \infty]$ is an $\mathcal{M}$ -measurable function, we define its **integral** with respect to μ by:

$$\int f d\mu = \sup \left\{ \int \phi d\mu : 0 \leq \phi \leq f, \phi \text{ is a simple function} \right\}$$

Remark on the Case when f is Simple

An important check for consistency is ensuring that this new definition matches our original definition when f itself happens to be a simple function.

Proposition. *If f is a simple function in L^+ , then the two definitions of $\int f d\mu$ agree.*

Proof. Let $f \in L^+$ be a simple function. Let $I_{\text{old}}(f)$ denote the integral of f calculated using its standard representation, and let $I_{\text{new}}(f)$ denote the value given by the supremum formula above:

$$I_{\text{new}}(f) = \sup \left\{ \int \phi d\mu : 0 \leq \phi \leq f, \phi \text{ is simple} \right\}$$

To establish complete clarity for students, we show that $I_{\text{new}}(f) = I_{\text{old}}(f)$ by proving both inequalities:

Step 1: Proving $I_{\text{old}}(f) \leq I_{\text{new}}(f)$: The set over which the supremum is taken consists of all simple functions ϕ that satisfy the constraint $0 \leq \phi \leq f$. Since f is assumed to be a simple function, and it trivially satisfies $0 \leq f \leq f$, the function f is itself a member of this family of functions.

Because the supremum of a set is greater than or equal to any individual element belonging to that set, evaluating the set at $\phi = f$ gives:

$$I_{\text{new}}(f) \geq I_{\text{old}}(f)$$

Step 2: Proving $I_{\text{new}}(f) \leq I_{\text{old}}(f)$: By Proposition 2.13c, the simple integral preserves order: if ϕ and f are simple functions and $\phi \leq f$, then $\int \phi d\mu \leq \int f d\mu$.

Therefore, $I_{\text{old}}(f)$ acts as an upper bound for every single numerical value in the set $\{\int \phi d\mu : 0 \leq \phi \leq f, \phi \text{ simple}\}$. Since the supremum is the **least** upper bound, it cannot exceed this value:

$$I_{\text{new}}(f) \leq I_{\text{old}}(f)$$

Since both $I_{\text{old}}(f) \leq I_{\text{new}}(f)$ and $I_{\text{new}}(f) \leq I_{\text{old}}(f)$ are true, we conclude that the two definitions agree seamlessly. $\square$

Exercise 4. Let $f, g \in L^+$ and let $c \in [0, \infty)$. Show that:

$$1. \int f \leq \int g \text{ whenever } f \leq g.$$

$$2. \int cf = c \int f.$$

Proof. Recall that for any general nonnegative measurable function $h \in L^+$, its integral is defined by the supremum over all lower simple functions:

$$\int h d\mu = \sup \left\{ \int \phi d\mu : 0 \leq \phi \leq h, \phi \text{ is simple} \right\}$$

Proof of Part 1 (Monotonicity):

Let S_f and S_g denote the families of simple functions bounded above by f and g respectively:

$$S_f = \{\phi : 0 \leq \phi \leq f, \phi \text{ is simple}\}$$

$$S_g = \{\phi : 0 \leq \phi \leq g, \phi \text{ is simple}\}$$

Suppose $\phi \in S_f$. By definition, ϕ is a simple function satisfying $0 \leq \phi(x) \leq f(x)$ for all $x \in X$. We are given the hypothesis that $f(x) \leq g(x)$ for all $x \in X$. By the transitive property of real inequalities, since $\phi(x) \leq f(x)$ and $f(x) \leq g(x)$, it must hold that:

$$\phi(x) \leq g(x) \quad \text{for all } x \in X$$

Therefore, ϕ automatically satisfies the conditions to belong to S_g . This directly establishes the set containment:

$$S_f \subseteq S_g$$

In set theory and real analysis, if a set of real numbers A is a subset of a set B ($A \subseteq B$), then the supremum of the larger set B must be greater than or equal to the supremum of the smaller set A . Taking the supremum of the integrals over both collections gives:

$$\sup_{\phi \in S_f} \int \phi \, d\mu \leq \sup_{\phi \in S_g} \int \phi \, d\mu$$

By the definition of the integral for general nonnegative functions, this inequality is exactly:

$$\int f \, d\mu \leq \int g \, d\mu$$

Proof of Part 2 (Homogeneity):

We analyze this property by dividing it into two cases based on the value of c .

Case 2a: $c = 0$.

If $c = 0$, the function $cf = 0 \cdot f$ is identically the zero function 0. The only non-negative simple function satisfying $0 \leq \phi \leq 0$ is $\phi = 0$. Thus, its supremum definition yields:

$$\int 0 \cdot f \, d\mu = \sup\{0\} = 0$$

On the other hand, multiplying the scalar constant gives $c \int f \, d\mu = 0 \cdot \int f \, d\mu$. Under standard measure-theoretic conventions, even if $\int f \, d\mu = \infty$, the product evaluates via $0 \cdot \infty = 0$. Thus, the identity holds perfectly when $c = 0$.

Case 2b: $c > 0$.

When c is strictly positive, we establish equality by demonstrating both directions of the inequality.

Step 1: Proving $\int cf \, d\mu \geq c \int f \, d\mu$.

Let ϕ be any simple function satisfying $0 \leq \phi \leq f$. Multiplying this pointwise inequality by the positive real constant $c > 0$ preserves the inequality direction:

$$0 \leq c\phi \leq cf$$

Since ϕ is simple, the scaled function $c\phi$ is also a simple function. Because $c\phi$ is a valid lower simple function for cf , the definition of the general integral as a supremum implies:

$$\int cf \, d\mu \geq \int c\phi \, d\mu$$

By Proposition 2.13a (homogeneity for simple functions), we have $\int c\phi \, d\mu = c \int \phi \, d\mu$. Substituting this yields:

$$\int cf \, d\mu \geq c \int \phi \, d\mu \implies \frac{1}{c} \int cf \, d\mu \geq \int \phi \, d\mu$$

Since this upper bound holds uniformly for every simple function $\phi \leq f$, it must also bound their least upper bound:

$$\frac{1}{c} \int cf \, d\mu \geq \sup_{0 \leq \phi \leq f} \int \phi \, d\mu = \int f \, d\mu$$

Multiplying both sides by c gives our first directional inequality:

$$\int cf \, d\mu \geq c \int f \, d\mu$$

Step 2: Proving $\int cf \, d\mu \leq c \int f \, d\mu$.

Let ψ be any simple function satisfying $0 \leq \psi \leq cf$. Because $c > 0$, we can divide the pointwise inequality by c to obtain:

$$0 \leq \frac{1}{c}\psi \leq f$$

The function $\frac{1}{c}\psi$ is simple because it is a scalar multiple of the simple function ψ . Therefore, $\frac{1}{c}\psi$ belongs to the family of lower simple functions for f . By definition of the supremum:

$$\int f \, d\mu \geq \int \left(\frac{1}{c}\psi\right) \, d\mu$$

Applying simple function homogeneity (Proposition 2.13a) shows $\int (\frac{1}{c}\psi) \, d\mu = \frac{1}{c} \int \psi \, d\mu$:

$$\int f \, d\mu \geq \frac{1}{c} \int \psi \, d\mu \implies c \int f \, d\mu \geq \int \psi \, d\mu$$

Because the constant value $c \int f \, d\mu$ acts as a uniform upper bound for all choices of simple functions $\psi \leq cf$, it must be greater than or equal to their supremum:

$$c \int f \, d\mu \geq \sup_{0 \leq \psi \leq cf} \int \psi \, d\mu = \int cf \, d\mu$$

Combining the conclusions of Step 1 and Step 2 establishes that $\int cf = c \int f$ holds perfectly for all parameters. $\square$